BLADE: Bilevel Low-rank Augmented-Lagrangian Driven Erasure for LLM Unlearning

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Abstract

Large language models memorize training data, creating legal, privacy, and safety risks that call for targeted knowledge removal without degrading general capabilities. Existing unlearning methods use unbounded forget objectives that destroy model coherence, or fixed-weight loss balancing that cannot adapt as the difficulty of retain preservation shifts during training.

We propose BLADE, a bilevel optimization framework with two key innovations. First, we introduce *clamped entropy*, a forget objective that maximizes per-token output entropy only up to a vocabulary-invariant threshold $\tau \cdot H_{\max}$, after which the gradient is exactly zero—giving a bounded, self-stabilizing loss where forgotten tokens automatically drop out of optimization. Second, we formulate retain preservation as an *inequality-constrained* bilevel problem: the inner loop repairs retain performance via fast SGD steps, while the outer loop drives forgetting subject to an augmented Lagrangian whose asymmetric dual update permanently calibrates the tradeoff from a single constraint violation. All updates are confined to LoRA adapters.

We evaluate on four benchmarks spanning two model families and three scales: TOFU (Llama-3.2-1B/3B, three forget splits, 5 seeds), MUSE Books and News (Llama-2-7B), and KnowUndo (Llama-2-7B-chat, copyright and privacy domains). On TOFU 1B forget01, BLADE achieves HM=0.808 vs. the strongest baseline at 0.690 (+0.118), winning 4 of 6 settings on automated metrics and all 6 on an LLM judge that detects residual leakage missed by token-level statistics. On MUSE Books, BLADE reaches HM=0.823, surpassing both the retrained model (0.739) and the best baseline (0.741). On KnowUndo, BLADE achieves the best harmonic mean on both copyright (0.479) and privacy (0.601) domains. A complete ablation study confirms that each component is independently essential, and training dynamics analysis reveals a consistent three-phase convergence pattern, in contrast to baselines that exhibit uncontrolled retain oscillations.

1 Introduction

Large language models (LLMs) memorize substantial portions of their training data (Carlini et al., 2021). This creates risks including copyright infringement, privacy violations, and retention of hazardous knowledge. *Machine unlearning* aims to remove specific knowledge from a trained model without full retraining, which is computationally prohibitive for models with billions of parameters trained on trillions of tokens.

The central challenge in LLM unlearning is the *forget–retain tradeoff*: aggressively removing memorized content perturbs shared representations and degrades the model's general capabilities. Existing methods handle this tradeoff through loss balancing, combining a forget loss with a retain loss using fixed weights (Liu et al., 2025; Zhang et al., 2024). But fixed coefficients cannot adapt to the changing dynamics of unlearning, since the difficulty of retain preservation shifts as forgetting progresses.

We identify three failure modes in current approaches:

1. **Unbounded forgetting.** Gradient ascent (Jang et al., 2023) and logit-based objectives produce arbitrarily large gradients on highly memorized tokens. This causes catastrophic utility loss even with retain regularization.

2. **Static tradeoff.** Fixed-weight combinations of forget and retain losses (GradDiff,

NPO) cannot adapt when retain performance suddenly degrades during training.

3. **Over-forgetting.** Unclamped entropy maximization (Yuan et al., 2025) and gradient ascent keep pushing already-forgotten tokens. This wastes gradient budget and destabilizes the model.

We address these with BLADE, a bilevel optimization framework whose two main contributions are a novel forget objective and a novel bilevel constrained formulation:

1. **Clamped entropy loss** (Section 3.4): we maximize per-token output entropy up to a vocabulary-invariant threshold $\tau \cdot H_{\max}$, where $H_{\max} = \log V$. The threshold τ has a direct information-theoretic interpretation: a token is "forgotten" when its distribution reaches τ fraction of maximum uncertainty, regardless of vocabulary size. The $\max(0, \cdot)$ clamp creates a hard deadzone—once forgotten, a token contributes exactly zero gradient, bounding the loss in $[0, \tau \cdot H_{\max}]$ and making optimization self-stabilizing without any scheduling. No prior unlearning method provides this per-token stopping criterion.

2. **Inequality-constrained bilevel formulation** (Section 3.3, Section 3.5): we separate the retain and forget objectives into inner and outer loops of a bilevel optimization—the inner loop rapidly repairs retain performance via K SGD steps, while the outer loop drives forgetting subject to a hard inequality constraint $\mathcal{L}_{\text{ret}} \leq \varepsilon$. The constraint is enforced via an augmented Lagrangian with an asymmetric dual update that ratchets up protection on violations but decays slowly when satisfied. This "repair first, forget carefully" structure is the inverse of prior bilevel unlearning methods that put forgetting in the inner loop.

3. **LoRA parameterization** (Section 3.2): we confine all updates to low-rank adapters, which limits how much the unlearning update can affect other weights.

We evaluate BLADE on four benchmarks spanning two model families (Llama-2, Llama-3.2) and three scales (1B, 3B, 7B): TOFU (Maini et al., 2024) across two model scales and three forget splits (5 seeds each), MUSE Books and News (Shi et al., 2025) with Llama-2-7B, and KnowUndo (Tian et al., 2024) with Llama-2-7B-chat on copyright and privacy domains. On TOFU 1B forget01, BLADE achieves HM=0.808 versus the strongest baseline PDU (0.690), an improvement of +0.118. BLADE wins 4 of 6 TOFU settings on automated HM and all 6 on an LLM judge, with the largest gains on small forget sets where the forget–retain boundary is tightest. The LLM judge reveals near-zero forget leakage for BLADE (FL=0.015/2.0 on forget01) while the strongest baseline still leaks substantially (FL=0.855). On MUSE Books, BLADE achieves HM=0.823, the highest among all tested methods. On KnowUndo, BLADE achieves the best harmonic mean on both copyright (0.479 vs. PDU 0.456) and privacy (0.601 vs. PDU 0.552) domains, with the LLM judge confirming the highest retain accuracy. On MUSE News—a harder distributed multi-document setting—BLADE is competitive (HM=0.542) though PDU leads (0.581), highlighting the challenge of heterogeneous forget sets.

Beyond end-point metrics, we provide a complete ablation study and dynamics-level analysis showing that BLADE's training consistently follows a three-phase pattern—warm-up, spike-and-ratchet, smooth convergence—while baselines exhibit recurring retain oscillations with no self-correcting mechanism. The ablation confirms that each component (clamped entropy, bilevel structure, ALM constraint, LoRA parameterization) is necessary: removing any one causes catastrophic failure or significant degradation.

2 Related Work

2.1 Machine Unlearning for LLMs

The concept of machine unlearning was first formalized by Cao and Yang (2015), who proposed efficient data removal from learned models without full retraining. Bourtoule et al. (2021) later introduced SISA training, which partitions data during training to make unlearning tractable. However, these approaches target classification models and do not apply to large autoregressive language models.

For LLMs, gradient ascent (GA) is the most direct approach: maximize the loss on the forget set to push the model away from memorized content (Jang et al., 2023). GA is unbounded, the loss grows without limit, and it rapidly destroys model coherence on all inputs. Yao et al. (2024) proposed GradDiff, combining gradient ascent on forget data with cross-entropy on retain data. But retain performance still degrades because the forget loss drives weights in destructive directions.

A preference-optimization line of work began with NPO (Zhang et al., 2024), which adapts DPO (Rafailov et al., 2023) by treating forget samples as negative preferences. NPO collapses exponentially slower than GA, but it requires a frozen reference model in memory and its objective saturates instead of converging to a well-defined target. Fan et al. (2025) address the reference-model cost with SimNPO, dropping the reference model entirely while remaining competitive on TOFU and MUSE. Jin et al. (2025) take a multi-task view with NGDiff, normalizing the gradient difference between forget and retain objectives so that one does not dominate the other.

At the representation level, Li et al. (2024) proposed RMU as part of the WMDP benchmark. RMU steers intermediate representations toward random targets for forget data while anchoring retain representations. It works well in their benchmark but lacks the optimization-theoretic grounding of loss-based methods.

What all these approaches have in common is the absence of a *bounded, self-stabilizing* forget objective with a per-token stopping criterion. GA is unbounded and diverges. NPO's adaptive weighting damps asymptotically but never reaches exactly zero—it still applies gradient to well-forgotten sequences. Naïve entropy maximization (Yuan et al., 2025) pushes *all* tokens toward uniform indefinitely, with no concept of "sufficiently forgotten." The inverted hinge loss of LoKU (Cha et al., 2025) is bounded but operates on pairwise token margins, not distributional entropy, and provides no tunable threshold.

Our clamped entropy loss is qualitatively different: it defines a vocabulary-invariant sufficiency threshold ($\tau \cdot \log V$) and produces *exactly zero* gradient once a token crosses it. This hard deadzone means forgotten tokens permanently drop out of optimization—the loss is bounded by construction, the gradient norm shrinks monotonically, and no reference model is needed.

2.2 Unlearning Benchmarks and Evaluation

Evaluating unlearning is arguably as hard as the unlearning itself. Maini et al. (2024) introduced TOFU, a benchmark built around 200 synthetic author biographies with associated QA pairs. Since the data is fictitious, a gold-standard retrained model naturally exists. They evaluate using truth ratios, ROUGE scores, and membership inference attacks, and find that no tested baseline achieves truly effective unlearning.

Shi et al. (2025) proposed MUSE, which defines six desirable properties for an unlearned model: no verbatim memorization, no knowledge memorization, no privacy leakage, utility preservation, scalability, and sustainability. They evaluate on two realistic corpora (news articles and Harry Potter books) with Llama-2-7b as the base model. Their results show that while some methods reduce memorization, most fail on privacy leakage or severely degrade utility. The WMDP benchmark (Li et al., 2024) takes a safety-oriented perspective, measuring hazardous knowledge in biosecurity and cybersecurity domains.

We evaluate on both TOFU and MUSE because they provide complementary views: TOFU tests unlearning of factual associations in a controlled setting, while MUSE tests removal of naturalistic memorized content at larger scale.

2.3 Parameter-Efficient Unlearning

LoRA (Hu et al., 2022) has become the standard method for parameter-efficient adaptation of LLMs. It factorizes weight updates into low-rank matrices, reducing trainable parameters and memory by orders of magnitude. Several recent works apply LoRA to unlearning. Cha et al. (2025) proposed LoKU, which addresses the poor forget–retain trade-off when combining GA with LoRA by introducing an inverted hinge loss and Fisher-informed LoRA initialization. Jia et al. (2024a) proposed WAGLE, which assigns importance scores to model weights and applies strategic masking for modular unlearning.

In BLADE, we use LoRA not just for memory efficiency but as a *structural constraint*: the low-rank bottleneck limits the expressivity of the unlearning update, preventing the catastrophic weight changes that happen with full-parameter methods. This complements our bilevel formulation. The inner loop protects retain performance within a small parameter subspace, while the outer loop drives forgetting through the same bottleneck.

2.4 Bilevel Optimization

Bilevel optimization has a long history in operations research (Bertsekas, 1999) and is increasingly used in machine learning. MAML (Finn et al., 2017) optimizes for fast adaptation in the outer loop while performing task-specific fine-tuning in the inner loop. Franceschi et al. (2018) provided a unified bilevel framework for hyperparameter optimization and meta-learning. Computing exact hypergradients requires second-order information, so various approximations exist: implicit differentiation (Lorraine et al., 2020), Neumann series for inverse Hessian approximation (Ji et al., 2021), and first-order approaches that avoid Hessian computation entirely.

The augmented Lagrangian method (ALM) (Bertsekas, 1999; Nocedal and Wright, 2006) maintains dual variables alongside the primal objective for constrained optimization. Unlike penalty methods that require driving the penalty to infinity, ALM can achieve exact constraint satisfaction with finite parameters through adaptive dual updates. ALM is well-suited for our setting because retain performance must be strictly enforced as a constraint, not loosely balanced as a loss term.

The most closely related works along the constrained optimization axis are BLUR (Reisizadeh et al., 2026), PDU (Entesari et al., 2025), and Cheng et al. (2026). BLUR uses the retain loss as the lower-level objective and the forget loss as the upper-level one, optimizing through implicit differentiation. PDU uses a standard Lagrangian (linear penalty only, no quadratic term) with a logit-margin forget loss and symmetric dual updates. Cheng et al. (2026) use an augmented Lagrangian for unlearning, but with an *equality* constraint ($L_{\text{ret}} = L_{\text{ret}}(\boldsymbol{\theta}_0)$, penalty always active), a flat single-level optimizer (not bilevel), and evaluation only on vision models (ResNet-18, ViT-B) and classification tasks.

Our approach differs from all three in structural ways: (a) we enforce an *inequality* constraint $L_{\text{ret}} \leq \varepsilon$ where the quadratic penalty activates only on violation—the model is never penalized for being better than baseline, unlike equality-constrained formulations; (b) we use an *asymmetric* dual update (fast λ increase on violation, $10\times$ slower decay when satisfied), producing one-shot ratchet behavior rather than the recurring oscillations of symmetric updates; (c) the retain objective is nested in the *inner loop* of a bilevel problem, creating a "repair first, forget carefully" dynamic that inverts SCRUB's inner-loop forgetting; and (d) our forget objective is clamped entropy, a bounded loss with a hard per-token deadzone—in contrast to PDU's unbounded logit margin and the unbounded gradient ascent used in BLUR. Jia et al. (2024b) also explored second-order optimization for unlearning through Newton-step approximations via influence functions (Koh and Liang, 2017).

Our earlier attempt at bilevel unlearning used binary sparsity masks to select which parameters to update. LoRA's continuous low-rank structure turned out to be both more stable and more effective, which motivated the current BLADE design.

3 Method

We formulate LLM unlearning as constrained bilevel optimization. The inner loop preserves retain performance and the outer loop drives forgetting subject to an augmented Lagrangian (ALM) constraint. We confine all updates to LoRA adapters attached to a frozen base model.

3.1 Problem Setup

Let M be a pretrained LLM with parameters $\boldsymbol{\theta}_0$, $\mathcal{D}_{\text{forget}}$ the data to unlearn, and $\mathcal{D}_{\text{retain}}$ a set of data the model should preserve. We seek parameters $\boldsymbol{\theta}$ such that (i) the model's output distribution on $\mathcal{D}_{\text{forget}}$ is high-entropy (memorized content is forgotten) and (ii) the cross-entropy loss on $\mathcal{D}_{\text{retain}}$ remains below a threshold ε.

3.2 LoRA Parameterization

Rather than modifying the full weight matrices, we attach LoRA adapters (Hu et al., 2022) to all projection matrices ($q/k/v/o\_\mathrm{proj}$, gate/up/down_proj):

$$W' = W_0 + \tfrac{\alpha}{r}\,BA, \quad B\in\mathbb{R}^{d\times r},\ A\in\mathbb{R}^{r\times d} \tag{1}$$

where $r \ll d$ is the rank and α/r is a scaling factor. With $r = 16$ on a 7B model, only $\sim$0.08% of parameters are trainable. This has two practical benefits for unlearning: (a) the low-rank subspace limits the expressivity of the update and prevents catastrophic weight drift, and (b) disabling adapters instantly recovers the exact original model, which simplifies debugging and rollback.

3.3 Bilevel Formulation

We decompose the unlearning problem into two nested optimization levels with a deliberate assignment: *retain in the inner loop, forget in the outer loop.*

Inner loop (retain preservation). For each outer step, run K SGD steps on the retain set:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_{\text{in}}\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}};\mathcal{B}_{\text{ret}}) \tag{2}$$

where $\mathcal{B}_{\text{ret}} \sim \mathcal{D}_{\text{retain}}$ is a mini-batch and $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss. The inner loop uses a fast learning rate (η_{in} = 2e-4) to rapidly restore any retain damage caused by the previous outer step.

Outer loop (constrained forgetting). The outer objective combines the forget loss with an inequality constraint on retain performance enforced via an augmented Lagrangian:

$$\mathcal{L}_{\text{outer}} = \mathcal{L}_{\text{fgt}} + \lambda\, r + \tfrac{\rho}{2}[\max(0,r)]^2,\ r = L_{\text{ret}} - \varepsilon \tag{3}$$

where $L_{\text{ret}} = \mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}};\mathcal{B}'_{\text{ret}})$ is evaluated on a *fresh* retain batch (separate from the inner loop), λ is the Lagrange multiplier, ρ is the quadratic penalty coefficient, and ε is the retain budget. The outer loop uses a slow learning rate (η_{out} = 3e-5), ensuring that each forgetting step is small enough for the inner loop to repair.

Repair-first principle. This assignment—retain inside, forget outside—inverts prior bilevel unlearning approaches (e.g., SCRUB (Kurmanji et al., 2023) and BLUR (Reisizadeh et al., 2026) put forgetting in the inner loop and retain repair in the outer loop). Our design reflects a key insight: forgetting is the *hard* direction that must be carefully metered, while retain repair is a *fast* operation that converges in few steps. The speed asymmetry ($\eta_{\text{in}}/\eta_{\text{out}} \approx$ 7–10×) guarantees that the model enters each outer step from a state with good retain performance, preventing the catastrophic retain collapse that occurs when aggressive forgetting outpaces repair.

3.4 Clamped Entropy Loss

Entropy maximization as a forget objective is well-known (Yuan et al., 2025), but unclamped formulations have a fundamental problem: they apply nonzero gradients to every token at every step, even those whose output distribution is already near-uniform. This produces unnecessarily large outer perturbations that destabilize bilevel optimization. We propose a qualitatively different objective—*clamped entropy*—that defines when a token is "sufficiently forgotten" and stops optimizing it:

$$\mathcal{L}_{\text{forget}} = \frac{1}{T}\sum_{t=1}^{T}\max\bigl(0,\ \tau\cdot H_{\max} - H(p_t)\bigr) \tag{4}$$

where $p_t = \mathrm{softmax}(z_t)$ is the output distribution at position t, $H(p_t) = -\sum_v p_t(v)\log p_t(v)$ is its Shannon entropy, $H_{\max} = \log V$ is the entropy of a uniform distribution over vocabulary V, and $\tau \in (0, 1)$ is the target fraction.

Vocabulary-invariant threshold. The normalization by $H_{\max} = \log V$ makes τ interpretable across models: $\tau = 0.7$ means "70% of maximum possible uncertainty," whether the vocabulary has 32K tokens (Llama-2) or 128K tokens (Llama-3). Without this normalization, a fixed entropy threshold would need to be re-tuned for each tokenizer.

Hard deadzone. The $\max(0,\cdot)$ clamp is the critical distinction from prior entropy-based objectives. Once $H(p_t) \geq \tau\cdot H_{\max}$, the gradient contribution of token t is *exactly zero*—not approximately small, not asymptotically vanishing, but identically zero. Tokens "drop out" of the loss as they are forgotten, creating three properties critical for stable bilevel unlearning:

- **Bounded loss**: $\mathcal{L}_{\text{forget}} \in [0, \tau \cdot H_{\max}]$. The outer perturbation magnitude is bounded by construction (Proposition 1), so the inner loop's K steps reliably repair retain damage regardless of training stage.
- **Self-stabilizing**: the number of "active" tokens (those with $H(p_t) < \tau \cdot H_{\max}$) decreases monotonically during successful training. The effective gradient norm shrinks automatically, naturally decelerating the outer loop without learning rate scheduling.
- **Reference-free**: requires only a single forward pass through the current model, unlike DPO-style losses (NPO, SimNPO) that need a frozen reference model in memory.

In contrast, unclamped entropy maximization (Yuan et al., 2025) applies gradient pressure to *all* tokens at *all* steps—including those already at 99% of $H_{\max}$—wasting optimization budget and creating gratuitous perturbations to shared representations. Gradient ascent (Jang et al., 2023) is even worse: its loss is unbounded ($-\log p \to \infty$ as $p \to 0$), so a single highly-memorized token can dominate the entire gradient (see Proposition 1c).

3.5 Augmented Lagrangian with Asymmetric Dual Update

The Lagrange multiplier λ controls how much weight retain preservation gets in the outer objective. Instead of tuning λ as a fixed hyperparameter, we learn it via a dual update rule.

A critical design choice is the *type* of constraint. Standard ALM for equality constraints uses a symmetric update and a penalty that is always active (Bertsekas, 1999). We instead enforce an *inequality* constraint $L_{\text{ret}} \leq \varepsilon$: the quadratic penalty $\frac{\rho}{2}[\max(0, r)]^2$ activates *only when violated* (Eq. 3). This means the model is never penalized for achieving retain loss *below* ε—it is free to improve on retain as long as it doesn't exceed the budget. On top of this, we introduce an *asymmetric* dual update:

$$\lambda \leftarrow \begin{cases} \lambda + \rho \cdot r & \text{if } r > 0 \ \ \text{(violation)} \\ \lambda + 0.1\rho \cdot r & \text{if } r \leq 0 \ \ \text{(satisfaction)} \end{cases} \tag{5}$$

The 10× slower decay means that once a retain violation drives λ up, the system permanently increases retain protection. Without this asymmetry, symmetric updates let λ drop back between spikes, causing repeated forget–retain oscillation cycles.

Algorithm 1 BLADE: Bilevel Augmented Lagrangian Unlearning

Require: Model M with frozen weights $\boldsymbol{\theta}_0$, $\mathcal{D}_{\text{forget}}$, $\mathcal{D}_{\text{retain}}$
Require: Hyperparams: K, η_{in}, η_{out}, ε_{mul}, ρ, λ_0, α, τ, T
1: Attach LoRA adapters to M; $\boldsymbol{\theta} \leftarrow$ LoRA params
2: $\lambda \leftarrow \lambda_0$
3: $\varepsilon \leftarrow \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}_{\text{CE}}^{(k)}(\mathcal{D}_{\text{retain}})$
4: **for** $t = 1$ **to** T **do**
5:   **Inner loop** (retain repair):
6:   **for** $k = 1$ **to** K **do**
7:     $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_{\text{in}} \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}}; \mathcal{B}_{\text{ret}})$
8:   **end for**
9:   **Forget loss** (clamped entropy):
10:   $\mathcal{L}_{\text{fgt}} \leftarrow \frac{1}{T} \sum_t \max(0,\ \tau \log V - H(p_t))$
11:   **Outer loss** (augmented Lagrangian):
12:   $r \leftarrow \mathcal{L}_{\text{CE}}(M_{\boldsymbol{\theta}}; \mathcal{B}'_{\text{ret}}) - \varepsilon$
13:   $\mathcal{L}_{\text{outer}} \leftarrow \mathcal{L}_{\text{fgt}} + \lambda r + \frac{\rho}{2}[\max(0, r)]^2$
14:   $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta_{\text{out}} \cdot \text{Adam}(\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{outer}})$
15:   **Dual update** (asymmetric ratchet):
16:   **if** $r > 0$ **then**
17:     $\lambda \leftarrow \lambda + \rho \cdot r$
18:   **else**
19:     $\lambda \leftarrow \lambda + \alpha\rho \cdot r$
20:   **end if**
21: **end for**
22: **return** M with trained LoRA adapters

Auto-epsilon. Instead of hand-tuning an absolute ε, we set it as a fraction of the model's initial retain loss. Before training, we run one inner loop and compute:

$$\varepsilon = \varepsilon_{\text{mul}} \cdot \frac{1}{K} \sum_{k=1}^{K} \mathcal{L}_{\text{CE}}^{(k)} \tag{6}$$

where $\mathcal{L}_{\text{CE}}^{(k)}$ is the retain loss at inner step k. This way, ε adapts automatically to the model and dataset.

3.6 Algorithm Summary

4 Results

4.1 Benchmarks and Metrics

We evaluate on four benchmarks: **TOFU** (Maini et al., 2024) (fictitious QA, Llama-3.2-

1B/3B, 3 forget splits), **MUSE Books/News** (Shi et al., 2025) (Harry Potter / BBC articles, Llama-2-7B), **KnowUndo** (Tian et al., 2024) (copyright/privacy, Llama-2-7B-chat). All results are 5-fold (seeds 42/123/456/789/1024) unless noted.

Composite scores:

- TOFU: HM = hmean(MU, 1−Prob, 1−ROUGE)
- MUSE: HM = hmean(1−fk, 1−vm, rk)
- KnowUndo: HM = hmean(1−fgt_R, ret_R, MMLU)
- LLM Judge: $\mathrm{HM_J}$ = hmean(1−FL/2, RA/2, rRQ/2)

LLM judge (Claude Opus 4.7) scores on 0–2 scale: Forget Leakage (FL↓), Retain Accuracy (RA↑), retain Response Quality (rRQ↑).

4.2 TOFU (Llama-3.2-1B-Instruct)

4.3 TOFU (Llama-3.2-3B-Instruct)

4.4 MUSE Books (Llama-2-7B)

4.5 MUSE News (Llama-2-7B)

4.6 KnowUndo (Llama-2-7B-chat)

All methods 5-fold except BLURNPO (OOM). MMLU evaluated once at seed=42, used as constant for HM computation.

4.7 Sustainability and Scalability (MUSE News)

Scalability: single-shot unlearning on progressively larger forget sets (1–4× the 889-sample standard). Sustainability: sequential unlearning where each step applies the method on a new 889-sample forget set starting from the previous checkpoint.

PDU collapses at 4× scale (HM=0.005, retain→0.002) and after 2 sequential steps (HM drops from 0.58 to 0.02). BLADE maintains stable performance: scalability HM stays 0.542–0.552 through 3×, and sustainability HM holds 0.542–0.525 through 4 sequential steps.

4.8 Ablation Study

Key findings: (1) ALM is essential—removing it collapses the model (A16, HM=0.233). (2) LoRA provides critical implicit regularization (A15, ret drops 0.658→0.306). (3) Clamped entropy is the only viable forget loss; GA (A4), NPO (A5), and logit margin (A17) all fail catastrophically even with full bilevel ALM. (4) Inner loop adds safety margin (+0.027 retain) but ALM alone works reasonably (A1). (5) Bilevel assignment matters: swapping (retain outer, forget inner) causes λ explosion (A18, vm=0.643).

4.9 Training Dynamics

BLADE follows a consistent three-phase pattern across all settings (Figure 1): (1) **Warm-up**: forget loss declines while retain stays near ε. (2) **Spike and ratchet**: accumulated forgetting disrupts shared representations; λ jumps from ∼1 to ∼2.2–4.5, permanently increasing retain protection. Final λ scales with difficulty: 2.4 (fgt01) → 3.6 (fgt05) → 4.5 (fgt10). (3) **Smooth convergence**: forget continues to zero, no further retain violations.

PDU's dynamics (Figure 2) contrast sharply: unbounded forget loss, recurring retain spikes (5+ on MUSE Books, wild oscillations on News), and no corrective mechanism. The oscillations directly cause worse final metrics.

5 Conclusion

We presented BLADE, a bilevel optimization framework for LLM unlearning built on two contributions. First, *clamped entropy*: a forget loss with a hard per-token deadzone at a vocabulary-invariant threshold $\tau \cdot \log V$, giving bounded and self-stabilizing forgetting where forgotten tokens produce exactly zero gradient. Second, an *inequality-constrained bilevel formulation* where retain repair runs in the inner loop and forgetting in the outer loop—the inverse of prior bilevel unlearning—with an augmented Lagrangian whose asymmetric dual update permanently calibrates the tradeoff from a single retain spike. All updates are confined to LoRA adapters, limiting how far the model can drift.

We evaluated BLADE on four benchmarks spanning two model families (Llama-2, Llama-3.2) and three scales (1B, 3B, 7B). On TOFU across three forget splits, two model scales, and 5 seeds, BLADE outperforms baselines on the majority of settings, with the largest gains on small forget sets (forget01 1B: +0.118 HM over PDU) where the tradeoff is hardest. LLM judge evaluation confirms near-zero forget leakage where baselines retain substantial residual knowledge. On MUSE Books, BLADE

Table 1: TOFU 1B full results (5 seeds). HM = hmean(MU, 1−Prob, 1−ROUGE). Best in **bold**.

Method	Forget 1% MU↑	Prob↓	ROUGE↓	HM↑	Forget 5% MU↑	Prob↓	ROUGE↓	HM↑	Forget 10% MU↑	Prob↓	ROUGE↓	HM↑
Gold	.597	.166	.414	.655	.599	.127	.383	.676	.591	.116	.379	.677
GradAscent	$.596_{\pm.00}$	$.477_{\pm.01}$	$.456_{\pm.01}$	$.553_{\pm.01}$	$.008_{\pm.01}$	$.003_{\pm.01}$	$.112_{\pm.08}$	$.022_{\pm.04}$	$.000_{\pm.00}$	$.000_{\pm.00}$	$.001_{\pm.00}$	$.000_{\pm.00}$
GradDiff	$.581_{\pm.01}$	$.421_{\pm.04}$	$.464_{\pm.02}$	$.564_{\pm.01}$	$.453_{\pm.00}$	$.073_{\pm.00}$	$.375_{\pm.01}$	$.614_{\pm.01}$	$.435_{\pm.00}$	$.047_{\pm.00}$	$.339_{\pm.01}$	$.617_{\pm.00}$
NPO	$.596_{\pm.00}$	$.474_{\pm.01}$	$.438_{\pm.02}$	$.560_{\pm.01}$	$.454_{\pm.01}$	$.245_{\pm.01}$	$.308_{\pm.01}$	$.603_{\pm.01}$	$.393_{\pm.03}$	$.213_{\pm.01}$	$.210_{\pm.01}$	$.590_{\pm.02}$
SimNPO	$.594_{\pm.00}$	$.858_{\pm.01}$	$.734_{\pm.02}$	$.240_{\pm.01}$	$.596_{\pm.00}$	$.848_{\pm.00}$	$.741_{\pm.01}$	$.248_{\pm.00}$	$.597_{\pm.00}$	$.842_{\pm.00}$	$.734_{\pm.01}$	$.255_{\pm.00}$
RMU	$.557_{\pm.00}$	$.414_{\pm.01}$	$.415_{\pm.00}$	$.576_{\pm.00}$	$.546_{\pm.00}$	$.369_{\pm.01}$	$.423_{\pm.01}$	$.583_{\pm.00}$	$.572_{\pm.00}$	$.106_{\pm.02}$	$.322_{\pm.01}$	$.691_{\pm.01}$
BLURNPO	$.598_{\pm.00}$	$.676_{\pm.01}$	$.588_{\pm.02}$	$.417_{\pm.01}$	$.518_{\pm.02}$	$.475_{\pm.07}$	$.407_{\pm.05}$	$.541_{\pm.04}$	$.089_{\pm.14}$	$.087_{\pm.04}$	$.253_{\pm.09}$	$.154_{\pm.20}$
PDU	$.602_{\pm.00}$	$.186_{\pm.01}$	$.313_{\pm.01}$	$.690_{\pm.01}$	$.588_{\pm.00}$	$.073_{\pm.02}$	$.214_{\pm.03}$	$.740_{\pm.01}$	$.592_{\pm.00}$	$.004_{\pm.00}$	$.065_{\pm.01}$	$.797_{\pm.00}$
BLADE	$.599_{\pm.00}$	**.002**$_{\pm.00}$	**.038**$_{\pm.00}$	**.808**$_{\pm.00}$	$.597_{\pm.00}$	**.015**$_{\pm.00}$	**.054**$_{\pm.01}$	**.800**$_{\pm.00}$	$.593_{\pm.01}$	**.009**$_{\pm.00}$	**.039**$_{\pm.01}$	**.803**$_{\pm.00}$

Table 2: LLM Judge on TOFU 1B (5 seeds, Claude Opus 4.7). Best in **bold**.

Method	Forget 1% FL↓	RA↑	rRQ↑	$\mathrm{HM_J}$↑	Forget 5% FL↓	RA↑	rRQ↑	$\mathrm{HM_J}$↑	Forget 10% FL↓	RA↑	rRQ↑	$\mathrm{HM_J}$↑
GradAscent	$1.11_{\pm.05}$	$1.66_{\pm.01}$	$1.99_{\pm.00}$	$.674_{\pm.02}$	$0.07_{\pm.07}$	$0.10_{\pm.09}$	$0.51_{\pm.43}$	$.117_{\pm.10}$	$0.00_{\pm.00}$	$0.00_{\pm.00}$	$0.00_{\pm.00}$	$.000_{\pm.00}$
GradDiff	$1.21_{\pm.07}$	$1.54_{\pm.04}$	$1.97_{\pm.01}$	$.620_{\pm.02}$	$0.65_{\pm.03}$	$1.01_{\pm.01}$	$1.68_{\pm.01}$	$.643_{\pm.00}$	$0.46_{\pm.01}$	$0.91_{\pm.02}$	$1.53_{\pm.04}$	$.625_{\pm.01}$
NPO	$1.05_{\pm.08}$	$1.67_{\pm.01}$	$1.99_{\pm.00}$	$.696_{\pm.03}$	$0.44_{\pm.02}$	$1.14_{\pm.05}$	$1.89_{\pm.02}$	$.731_{\pm.01}$	$0.44_{\pm.02}$	$0.94_{\pm.07}$	$1.82_{\pm.04}$	$.665_{\pm.03}$
SimNPO	$1.76_{\pm.03}$	$1.66_{\pm.01}$	$1.99_{\pm.00}$	$.288_{\pm.03}$	$1.58_{\pm.01}$	$1.69_{\pm.01}$	$1.99_{\pm.00}$	$.435_{\pm.01}$	$1.58_{\pm.02}$	$1.72_{\pm.02}$	$1.99_{\pm.00}$	$.435_{\pm.02}$
RMU	$0.81_{\pm.05}$	$1.39_{\pm.01}$	$1.93_{\pm.01}$	$.721_{\pm.01}$	$0.72_{\pm.04}$	$1.36_{\pm.02}$	$1.91_{\pm.01}$	$.736_{\pm.01}$	$0.29_{\pm.04}$	$1.50_{\pm.01}$	$1.98_{\pm.00}$	$.854_{\pm.01}$
BLURNPO	$1.40_{\pm.05}$	$1.67_{\pm.01}$	$1.99_{\pm.00}$	$.541_{\pm.03}$	$0.86_{\pm.08}$	$1.37_{\pm.06}$	$1.98_{\pm.01}$	$.709_{\pm.01}$	$0.26_{\pm.24}$	$0.36_{\pm.38}$	$0.73_{\pm.64}$	$.255_{\pm.21}$
PDU	$0.86_{\pm.06}$	$1.60_{\pm.01}$	$1.97_{\pm.00}$	$.746_{\pm.02}$	$0.37_{\pm.08}$	$1.56_{\pm.02}$	$1.96_{\pm.01}$	$.850_{\pm.01}$	$0.04_{\pm.01}$	$1.57_{\pm.01}$	$1.98_{\pm.01}$	$.906_{\pm.00}$
BLADE	**0.02**$_{\pm.02}$	$1.65_{\pm.01}$	$1.98_{\pm.01}$	**.929**$_{\pm.01}$	**0.07**$_{\pm.02}$	$1.64_{\pm.02}$	$1.98_{\pm.01}$	**.919**$_{\pm.00}$	**0.04**$_{\pm.02}$	$1.63_{\pm.02}$	$1.98_{\pm.00}$	**.919**$_{\pm.01}$

achieves HM=0.823, the highest among all tested methods including the gold retrained model. On KnowUndo, BLADE achieves the best harmonic mean on both copyright and privacy domains, demonstrating robustness to LoRA-fine-tuned target models and domain-specific unlearning tasks. On MUSE News, BLADE is competitive but does not dominate, highlighting that distributed multi-document unlearning remains an open challenge for entropy-based approaches.

A complete ablation study confirms that each component is necessary: removing the ALM causes model collapse (HM=0.233), removing LoRA causes catastrophic drift (HM=0.459), replacing clamped entropy with GA or NPO causes retain collapse or trivial satisfaction, and swapping the bilevel structure causes λ divergence. Training dynamics reveal a consistent three-phase convergence pattern across all scales and datasets, while baselines show recurring retain oscillations—providing direct evidence that the ALM formulation produces qualitatively different optimization behavior.

Limitations. BLADE has more hyperparameters than single-loss baselines (K, ρ, τ, ε_{mul}), although auto-epsilon and the observed robustness to ρ and K reduce the practical tuning surface. On MUSE News, a distributed multi-document setting, BLADE underperforms PDU, suggesting that a single global λ may be insufficient when forget difficulty varies substantially across documents. MUSE results use a single seed (5-fold evaluations are in progress). The bilevel structure adds computational cost ($K = 3$ inner steps per outer step), roughly 4× the wall time of single-loop methods.

Future work. Adaptive per-document or per-cluster constraint enforcement could address the MUSE News limitation. Evaluation on safety-critical unlearning (WMDP) and larger instruction-tuned models (8B+) is a natural next step. The interpretable three-phase dynamics suggest connections to phase transitions in learning that warrant theoretical investigation.

Table 3: TOFU 3B full results (5 seeds). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑	MU↑	Prob↓	ROUGE↓	HM↑
Gold	.663	.179	.409	.656	.659	.130	.388	.698	.661	.115	.382	.698
GradAscent	.668$_{\pm.00}$	.564$_{\pm.01}$	.525$_{\pm.01}$	.509$_{\pm.01}$	.481$_{\pm.02}$	.140$_{\pm.02}$	.333$_{\pm.01}$	.632$_{\pm.01}$	.000$_{\pm.00}$	.000$_{\pm.00}$	.003$_{\pm.00}$	.000$_{\pm.00}$
GradDiff	.659$_{\pm.00}$	.569$_{\pm.02}$	.578$_{\pm.03}$	.483$_{\pm.02}$	.566$_{\pm.01}$	.147$_{\pm.01}$	.395$_{\pm.01}$	.653$_{\pm.00}$	.532$_{\pm.02}$	.079$_{\pm.01}$	.361$_{\pm.02}$	.662$_{\pm.01}$
NPO	.668$_{\pm.00}$	.553$_{\pm.01}$	.494$_{\pm.01}$	.525$_{\pm.01}$	.542$_{\pm.01}$	.219$_{\pm.00}$	.359$_{\pm.01}$	.640$_{\pm.00}$	.549$_{\pm.01}$	.237$_{\pm.01}$	.353$_{\pm.06}$	.640$_{\pm.02}$
SimNPO	.656$_{\pm.00}$	.920$_{\pm.01}$	.830$_{\pm.02}$	.151$_{\pm.01}$	.657$_{\pm.00}$	.901$_{\pm.00}$	.818$_{\pm.01}$	.175$_{\pm.00}$	.652$_{\pm.00}$	.889$_{\pm.00}$	.808$_{\pm.00}$	.191$_{\pm.00}$
RMU	.657$_{\pm.00}$	.861$_{\pm.00}$	.715$_{\pm.01}$	.246$_{\pm.00}$	.644$_{\pm.00}$	.602$_{\pm.00}$	.534$_{\pm.00}$	.483$_{\pm.00}$	.647$_{\pm.00}$	.390$_{\pm.01}$	.472$_{\pm.00}$	.591$_{\pm.00}$
BLURNPO	.667$_{\pm.00}$	.822$_{\pm.02}$	.788$_{\pm.02}$	.253$_{\pm.02}$	.640$_{\pm.02}$	.683$_{\pm.08}$	.593$_{\pm.09}$	.412$_{\pm.08}$	.385$_{\pm.17}$	.245$_{\pm.16}$	.314$_{\pm.09}$	.502$_{\pm.10}$
PDU	.692$_{\pm.00}$	.168$_{\pm.01}$	.283$_{\pm.01}$	.742$_{\pm.00}$	.687$_{\pm.00}$	.009$_{\pm.00}$	.085$_{\pm.01}$	.843$_{\pm.00}$	.680$_{\pm.01}$	**.000**$_{\pm.00}$	.029$_{\pm.01}$	**.857**$_{\pm.00}$
BLADE	.654$_{\pm.00}$	**.000**$_{\pm.00}$	**.015**$_{\pm.01}$	**.846**$_{\pm.00}$	.655$_{\pm.00}$	**.007**$_{\pm.01}$	**.028**$_{\pm.02}$	.842$_{\pm.01}$	.647$_{\pm.00}$	.005$_{\pm.00}$	**.038**$_{\pm.01}$	.836$_{\pm.01}$

Table 4: LLM Judge on TOFU 3B (5 seeds, Claude Opus 4.7). Best in **bold**.

Method	Forget 1%				Forget 5%				Forget 10%			
	FL↓	RA↑	rRQ↑	HM$_J$↑	FL↓	RA↑	rRQ↑	HM$_J$↑	FL↓	RA↑	rRQ↑	HM$_J$↑
GradAscent	1.34$_{\pm.09}$	1.85$_{\pm.00}$	2.00$_{\pm.00}$	.587$_{\pm.05}$	0.69$_{\pm.04}$	1.45$_{\pm.02}$	1.98$_{\pm.01}$	.766$_{\pm.01}$	0.00$_{\pm.00}$	0.00$_{\pm.00}$	0.00$_{\pm.00}$	.000$_{\pm.00}$
GradDiff	1.50$_{\pm.07}$	1.82$_{\pm.02}$	2.00$_{\pm.00}$	.490$_{\pm.04}$	0.83$_{\pm.01}$	1.29$_{\pm.02}$	1.71$_{\pm.06}$	.677$_{\pm.01}$	0.66$_{\pm.03}$	1.16$_{\pm.06}$	1.43$_{\pm.20}$	.648$_{\pm.04}$
NPO	1.23$_{\pm.04}$	1.84$_{\pm.00}$	2.00$_{\pm.00}$	.640$_{\pm.02}$	0.74$_{\pm.04}$	1.55$_{\pm.01}$	2.00$_{\pm.00}$	.774$_{\pm.01}$	0.61$_{\pm.08}$	1.52$_{\pm.04}$	1.99$_{\pm.01}$	.796$_{\pm.01}$
SimNPO	1.83$_{\pm.05}$	1.84$_{\pm.01}$	2.00$_{\pm.00}$	.220$_{\pm.06}$	1.74$_{\pm.01}$	1.85$_{\pm.01}$	2.00$_{\pm.00}$	.305$_{\pm.01}$	1.71$_{\pm.01}$	1.86$_{\pm.01}$	2.00$_{\pm.00}$	.331$_{\pm.01}$
RMU	1.59$_{\pm.04}$	1.81$_{\pm.00}$	2.00$_{\pm.00}$	.432$_{\pm.03}$	1.11$_{\pm.03}$	1.70$_{\pm.01}$	1.99$_{\pm.00}$	.679$_{\pm.01}$	0.85$_{\pm.02}$	1.70$_{\pm.01}$	1.99$_{\pm.00}$	.765$_{\pm.01}$
BLURNPO	1.73$_{\pm.12}$	1.85$_{\pm.00}$	2.00$_{\pm.00}$	.308$_{\pm.11}$	1.40$_{\pm.15}$	1.71$_{\pm.05}$	2.00$_{\pm.00}$	.538$_{\pm.08}$	0.73$_{\pm.35}$	1.05$_{\pm.51}$	1.51$_{\pm.60}$	.550$_{\pm.16}$
PDU	0.62$_{\pm.04}$	1.72$_{\pm.01}$	1.97$_{\pm.00}$	.828$_{\pm.01}$	0.07$_{\pm.01}$	1.75$_{\pm.01}$	1.98$_{\pm.00}$	.941$_{\pm.00}$	**0.01**$_{\pm.01}$	1.73$_{\pm.03}$	1.95$_{\pm.04}$	.940$_{\pm.01}$
BLADE	**0.01**$_{\pm.01}$	1.85$_{\pm.00}$	2.00$_{\pm.00}$	**.970**$_{\pm.00}$	**0.02**$_{\pm.03}$	1.82$_{\pm.01}$	1.99$_{\pm.00}$	**.962**$_{\pm.00}$	0.03$_{\pm.03}$	1.84$_{\pm.01}$	1.99$_{\pm.00}$	**.965**$_{\pm.00}$

Table 5: MUSE Books (5-fold; †seed=42). HM = hmean(1−fk, 1−vm, rk). Best in **bold** (excl. Gold).

Method	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.303	.145	.687	.739
GradAscent	.000$_{\pm.00}$	.000$_{\pm.00}$	.000$_{\pm.00}$	.000$_{\pm.00}$
GradDiff	.000$_{\pm.00}$	.000$_{\pm.00}$	.004$_{\pm.00}$	.011$_{\pm.01}$
NPO	.303$_{\pm.02}$	.342$_{\pm.03}$	.574$_{\pm.02}$	.638$_{\pm.01}$
SimNPO	.238$_{\pm.02}$	.002$_{\pm.00}$	.600$_{\pm.01}$	.753$_{\pm.01}$
RMU	.210$_{\pm.00}$	.113$_{\pm.01}$	.598$_{\pm.01}$	.738$_{\pm.00}$
PDU	.139$_{\pm.01}$	.129$_{\pm.01}$	.372$_{\pm.01}$	.600$_{\pm.01}$
BLURNPO†	.175	.000	.527	.730
BLADE	**.089**$_{\pm.02}$	**.000**$_{\pm.00}$	**.638**$_{\pm.01}$	**.818**$_{\pm.01}$

†seed=42 only (BLURNPO OOM on 4×40GB for 5-fold).

Table 6: MUSE Books LLM judge (5-fold; †seed=42). Best in **bold**.

Method	FL↓	RA↑	rRQ↑	HM$_J$↑
GradAscent	0.00$_{\pm.00}$	0.00$_{\pm.00}$	0.00$_{\pm.00}$	.000$_{\pm.00}$
GradDiff	0.00$_{\pm.00}$	0.04$_{\pm.02}$	0.01$_{\pm.01}$	.010$_{\pm.01}$
NPO	0.91$_{\pm.03}$	1.28$_{\pm.03}$	1.61$_{\pm.01}$	.647$_{\pm.01}$
SimNPO	0.31$_{\pm.12}$	1.28$_{\pm.02}$	1.62$_{\pm.01}$	.753$_{\pm.02}$
RMU	0.32$_{\pm.02}$	1.41$_{\pm.02}$	1.68$_{\pm.01}$	.791$_{\pm.00}$
PDU	0.53$_{\pm.02}$	0.99$_{\pm.02}$	1.15$_{\pm.02}$	.586$_{\pm.01}$
BLURNPO†	0.21	1.09	1.47	.696
BLADE	**0.11**$_{\pm.02}$	**1.39**$_{\pm.01}$	**1.66**$_{\pm.02}$	**.810**$_{\pm.01}$

Table 7: MUSE News (5-fold; †seed=42). Best HM in **bold**.

Method	fk↓	vm↓	rk↑	HM↑
Gold (retrain)	.324	.204	.552	.660
GradDiff	.329$_{\pm.03}$	.043$_{\pm.02}$	.269$_{\pm.02}$	.479$_{\pm.02}$
NPO	.517$_{\pm.02}$	.274$_{\pm.02}$	.435$_{\pm.01}$	.522$_{\pm.01}$
SimNPO	.628$_{\pm.01}$	.542$_{\pm.01}$	.513$_{\pm.01}$	.440$_{\pm.00}$
RMU	.495$_{\pm.01}$	.267$_{\pm.01}$	.434$_{\pm.01}$	.531$_{\pm.00}$
PDU	.525$_{\pm.02}$	.092$_{\pm.07}$	.508$_{\pm.04}$	**.577**$_{\pm.01}$
BLURNPO†	.317	.181	.304	.448
BLADE †	.550	.173	.475	.542

A Theoretical Analysis

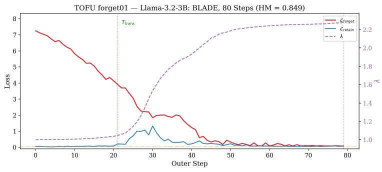
We provide formal justification for three key design choices in BLADE: (i) bounded loss and gradient control of the clamped entropy objective (§A.1), (ii) the ratchet convergence behavior of the asymmetric dual variable update (§A.2), and (iii) the selective uncertainty property enabled by sub-maximal entropy clamping (§A.3). Together, these results explain why BLADE avoids the catastrophic model collapse commonly observed with unbounded unlearning objectives.

A.1 Bounded Loss and Gradient of Clamped Entropy

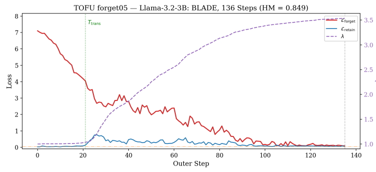
Let V denote the vocabulary size, $\tau \in (0,1)$ a clamping ratio, and $p_{\boldsymbol{\theta}}(\cdot \mid x_{<t})$ the next-token distribution produced by a language model with parameters $\boldsymbol{\theta}$. Define the per-token Shannon entropy $H_t(\boldsymbol{\theta}) = -\sum_{v=1}^{V} p_{\boldsymbol{\theta}}(v \mid x_{<t}) \log p_{\boldsymbol{\theta}}(v \mid x_{<t})$ and the clamped entropy loss for a single token as

$$\ell_t(\boldsymbol{\theta}) = \max\big(0,\ \tau \log V - H_t(\boldsymbol{\theta})\big). \quad (7)$$

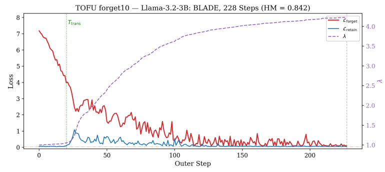
For a sequence of T tokens the forget loss is $\mathcal{L}_{\text{fgt}}(\boldsymbol{\theta}) = \frac{1}{T}\sum_{t=1}^{T} \ell_t(\boldsymbol{\theta})$.



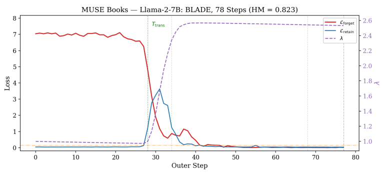
(a) BLADE– TOFU 3B fgt01



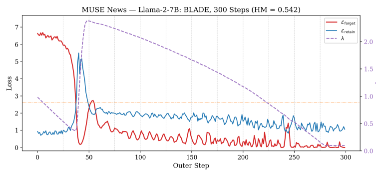
(b) BLADE– TOFU 3B fgt05



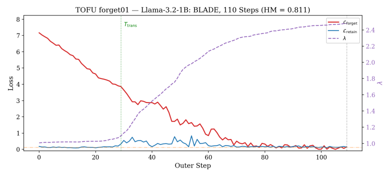
(c) BLADE– TOFU 3B fgt10



(d) BLADE– MUSE Books

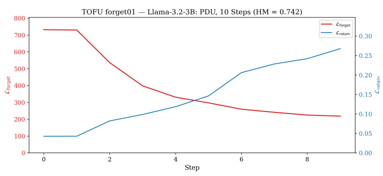


(e) BLADE– MUSE News

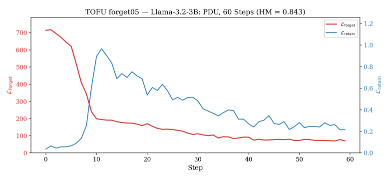


(f) BLADE– TOFU 1B fgt01

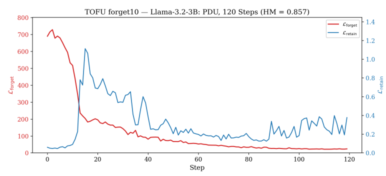
Figure 1: BLADE training dynamics across benchmarks. All settings show the same three-phase pattern: warm-up $\to$ spike-and-ratchet $\to$ smooth convergence. λ adapts to task difficulty (higher for larger forget sets).



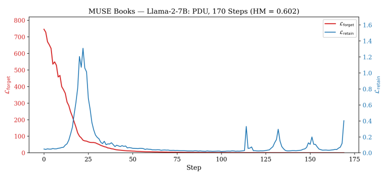
(a) PDU – TOFU 3B fgt01



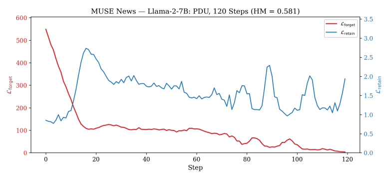
(b) PDU – TOFU 3B fgt05



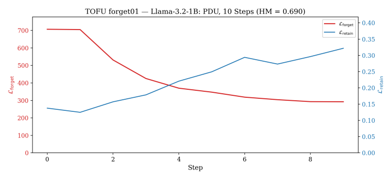
(c) PDU – TOFU 3B fgt10



(d) PDU – MUSE Books



(e) PDU – MUSE News



(f) PDU – TOFU 1B fgt01

Figure 2: PDU training dynamics. Unbounded logit-margin loss (starts at 700+) and symmetric dual updates produce recurring retain oscillations across all settings. No self-correcting mechanism exists.

Table 8: MUSE News LLM judge (5-fold; †seed=42). Best in **bold**.

Method	FL↓	RA↑	rRQ↑	HM$_J$↑
GradDiff	$0.58_{\pm.02}$	$0.79_{\pm.04}$	$1.16_{\pm.14}$	$.529_{\pm.03}$
NPO	$1.20_{\pm.03}$	$1.05_{\pm.02}$	$1.58_{\pm.01}$	$.531_{\pm.01}$
SimNPO	$1.49_{\pm.01}$	$1.24_{\pm.01}$	$1.72_{\pm.01}$	$.447_{\pm.01}$
RMU	$1.13_{\pm.01}$	$1.15_{\pm.02}$	$1.60_{\pm.01}$	$.565_{\pm.00}$
PDU	$0.87_{\pm.12}$	$1.18_{\pm.05}$	$1.64_{\pm.05}$	$\mathbf{.638}_{\pm.02}$
BLURNPO†	0.47	0.51	1.47	.455
BLADE †	0.85	0.88	1.72	.579

Proposition 1 (Bounded Loss and Gradient of Clamped Entropy). *Assume: (A1) the logit map $z_t : \mathbb{R}^d \to \mathbb{R}^V$ is G-Lipschitz, i.e., $\|\nabla_{\boldsymbol{\theta}} z_t(\boldsymbol{\theta})\|_F \le G$; and (A2) the logit magnitudes are bounded, $\|z_t(\boldsymbol{\theta})\|_\infty \le B$, for all tokens t and parameters $\boldsymbol{\theta}$. Then:*

(a) *The loss is bounded: $0 \le \ell_t(\boldsymbol{\theta}) \le \tau \log V$ for all $\boldsymbol{\theta}$.*

(b) *The per-token gradient norm satisfies*

$$\|\nabla_{\boldsymbol{\theta}} \ell_t(\boldsymbol{\theta})\| \le (2B + 2\log V)\, G. \qquad (8)$$

(c) *In contrast, the gradient ascent loss $\mathcal{L}_{\mathrm{GA}} = -\log p_{\boldsymbol{\theta}}(y_t \mid x_{<t})$ has unbounded loss values as $p_{\boldsymbol{\theta}}(y_t) \to 0$, and unclamped entropy maximization $\mathcal{L}_{\mathrm{ent}} = -H_t$ applies nonzero gradients even when H_t is already near $H_{\max}$.*

Proof. (a) Since $0 \le H_t \le \log V$ and $\tau \in (0, 1)$, the ReLU outputs a value in $[0, \tau \log V]$.

(b) In the active region ($H_t < \tau \log V$), the chain rule gives $\nabla_{\boldsymbol{\theta}} \ell_t = -\nabla_{\boldsymbol{\theta}} H_t$. We derive a component-wise bound on $\nabla_{z_t} H_t$ that avoids the $\sqrt{V}$ factor introduced by spectral-norm arguments. The v-th component of the entropy

Table 9: KnowUndo Copyright (5-fold; †seed=42). HM = hmean(1−fgt_R, ret_R, MMLU). Best in **bold**.

	Automated Metrics					LLM Judge (Haiku 3.5)			
Method	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	HM↑	FL↓	RA↑	rRQ↑	HM_J↑
GradAscent	$.002_{\pm.00}$	$.000_{\pm.00}$	$.002_{\pm.00}$	$.000_{\pm.00}$	—	$0.00_{\pm.00}$	$0.00_{\pm.00}$	$0.00_{\pm.00}$	—
GradDiff	$.002_{\pm.00}$	$.000_{\pm.00}$	$.642_{\pm.04}$	$.199_{\pm.01}$	$.348_{\pm.01}$	$0.06_{\pm.07}$	$0.96_{\pm.16}$	$0.84_{\pm.14}$	$.47_{\pm.06}$
NPO	$.678_{\pm.02}$	$.189_{\pm.01}$	$.763_{\pm.02}$	$.289_{\pm.00}$	$.433_{\pm.00}$	$1.37_{\pm.45}$	$1.77_{\pm.10}$	$1.86_{\pm.16}$	$.49_{\pm.15}$
SimNPO	$.303_{\pm.05}$	$.048_{\pm.04}$	$.788_{\pm.01}$	$.317_{\pm.00}$	$.461_{\pm.01}$	$0.29_{\pm.22}$	$1.76_{\pm.13}$	$1.85_{\pm.18}$	$.80_{\pm.04}$
RMU	$.435_{\pm.01}$	$.076_{\pm.00}$	$.763_{\pm.00}$	$.201_{\pm.01}$	$.347_{\pm.01}$	$0.42_{\pm.22}$	$1.12_{\pm.12}$	$1.42_{\pm.10}$	$.62_{\pm.03}$
PDU	$.045_{\pm.00}$	$.011_{\pm.00}$	$.836_{\pm.00}$	$.285_{\pm.01}$	$.442_{\pm.01}$	$0.08_{\pm.04}$	$1.61_{\pm.07}$	$1.58_{\pm.13}$	$.75_{\pm.03}$
BLURNPO†	.494	.023	.689	.140	.287	0.01	0.24	0.32	—
BLADE	$\mathbf{.189}_{\pm.14}$	$\mathbf{.014}_{\pm.01}$	$\mathbf{.840}_{\pm.01}$	$\mathbf{.340}_{\pm.00}$	$\mathbf{.482}_{\pm.00}$	$0.24_{\pm.19}$	$\mathbf{1.81}_{\pm.03}$	$\mathbf{1.89}_{\pm.13}$	$\mathbf{.82}_{\pm.03}$

Table 10: KnowUndo Privacy (5-fold; †seed=42). HM = hmean(1−fgt_R, ret_R, MMLU). Best in **bold**.

	Automated Metrics					LLM Judge (Haiku 3.5)			
Method	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	HM↑	FL↓	RA↑	rRQ↑	HM_J↑
GradAscent	$.017_{\pm.02}$	$.004_{\pm.01}$	$.012_{\pm.01}$	$.006_{\pm.01}$	—	$0.00_{\pm.00}$	$0.00_{\pm.00}$	$0.00_{\pm.00}$	—
GradDiff	$.049_{\pm.02}$	$.039_{\pm.02}$	$.708_{\pm.03}$	$.406_{\pm.02}$	$.516_{\pm.01}$	$0.04_{\pm.02}$	$1.08_{\pm.07}$	$1.28_{\pm.15}$	$.50_{\pm.04}$
NPO	$.646_{\pm.02}$	$.322_{\pm.02}$	$.790_{\pm.00}$	$.426_{\pm.01}$	$.496_{\pm.00}$	$0.99_{\pm.23}$	$1.40_{\pm.14}$	$1.97_{\pm.02}$	$.66_{\pm.04}$
SimNPO	$.516_{\pm.04}$	$.302_{\pm.03}$	$.889_{\pm.01}$	$.557_{\pm.01}$	$.546_{\pm.01}$	$0.79_{\pm.23}$	$1.55_{\pm.12}$	$1.89_{\pm.04}$	$.66_{\pm.02}$
RMU	$.568_{\pm.01}$	$.072_{\pm.00}$	$.661_{\pm.01}$	$.079_{\pm.01}$	$.182_{\pm.01}$	$0.13_{\pm.06}$	$0.17_{\pm.07}$	$0.52_{\pm.15}$	$.18_{\pm.07}$
PDU	$.055_{\pm.02}$	$.022_{\pm.01}$	$.813_{\pm.02}$	$.452_{\pm.02}$	$.545_{\pm.01}$	$0.05_{\pm.03}$	$1.25_{\pm.06}$	$1.53_{\pm.04}$	$.58_{\pm.02}$
BLURNPO†	.103	.038	.704	.362	.498	0.04	0.42	0.84	—
BLADE	$\mathbf{.194}_{\pm.03}$	$\mathbf{.075}_{\pm.03}$	$\mathbf{.930}_{\pm.02}$	$\mathbf{.623}_{\pm.03}$	$\mathbf{.602}_{\pm.01}$	$0.20_{\pm.12}$	$\mathbf{1.50}_{\pm.07}$	$\mathbf{1.74}_{\pm.09}$	$\mathbf{.66}_{\pm.03}$

Table 11: Scalability on MUSE News.

Scale	Method	fk↓	vm↓	rk↑	HM↑
1× (889)	PDU	.542	.065	.522	.581
	BLADE	.550	.173	.475	.542
2× (1778)	PDU	.495	.050	.467	.579
	BLADE	.504	.321	.500	.547
3× (2667)	PDU	.538	.024	.488	.573
	BLADE	.466	.366	.505	.552
4× (3554)	PDU	.321	.021	.002	.005
	BLADE		(running)		

Table 12: Sustainability on MUSE News (sequential unlearning steps).

Step	Method	fk↓	vm↓	rk↑	HM↑
1	PDU	.542	.065	.522	.580
	BLADE	.550	.173	.475	.542
2	PDU	.554	.007	.470	.558
	BLADE	.539	.212	.485	.545
3	PDU	.550	.001	.005	.016
	BLADE	.556	.230	.459	.523
4	PDU	.591	.003	.051	.130
	BLADE	.564	.222	.469	.525

gradient with respect to logits is:

$$\frac{\partial H_t}{\partial z_{t,v}} = -p_v\big(H_t + \log p_v\big),$$

which follows from the softmax Jacobian identity $\partial p_j/\partial z_{t,v} = p_j(\delta_{jv} - p_v)$ applied to $H_t = -\sum_v p_v \log p_v$. One can verify correctness at the uniform distribution: $p_v = 1/V$ gives $H_t = \log V$ and $\log p_v = -\log V$, so $H_t + \log p_v = 0$ and the gradient vanishes, as expected at the entropy maximum. Under assumption (A2), the softmax probabilities satisfy $p_v \geq e^{-2B}/V$ for all v, since $p_v = e^{z_v}/\sum_j e^{z_j} \geq e^{-B}/(Ve^B) = e^{-2B}/V$. This yields $|\log p_v| \leq 2B + \log V$ and $H_t \leq \log V$, so $|H_t + \log p_v| \leq H_t + |\log p_v| \leq 2B + 2\log V$.

Since $p_v^2 \leq p_v$ for $p_v \in [0, 1]$:

$$\begin{aligned}\|\nabla_{z_t} H_t\|^2 &= \sum_v p_v^2 (H_t + \log p_v)^2 \\ &\leq (2B + 2\log V)^2 \sum_v p_v = (2B + 2\log V)^2.\end{aligned}$$

By the chain rule and assumption (A1):

$$\begin{aligned}\|\nabla_{\boldsymbol{\theta}} \ell_t\| = \|\nabla_{\boldsymbol{\theta}} H_t\| &\leq \|\nabla_{z_t} H_t\| \cdot \|\nabla_{\boldsymbol{\theta}} z_t\|_F \\ &\leq (2B + 2\log V) \cdot G.\end{aligned}$$

In the inactive region ($H_t \geq \tau \log V$), $\nabla_{\boldsymbol{\theta}} \ell_t = \mathbf{0}$. Averaging over T tokens preserves the bound.

Table 13: Ablation on MUSE Books (seed=42). Each row changes one component from the full BLADE. LLM judge scores on 0–2 scale.

Configuration	fk↓	vm↓	ret↑	HM↑	Δret	$\mathrm{FL_J}$↓	$\mathrm{RA_J}$↑	$\mathrm{rRQ_J}$↑	$\mathrm{HM_J}$↑
Full BLADE	.110	.000	.658	**.823**	—	0.14	1.40	1.69	**.814**
$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
$\tau=1.0$ (unclamped entropy)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
No LoRA (full fine-tuning)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
ALM off ($\lambda=0$, $\rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
Swapped bilevel (forget inner)	.372	.643	.651	.506	−.007	—	—	—	—
GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
NPO forget loss	.457	.997	.662	.009	+.004	1.44	1.47	1.72	.495
Logit margin forget loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000

(c) For GA, $-\log p_{\boldsymbol{\theta}}(y_t) \to \infty$ as $p_{\boldsymbol{\theta}}(y_t) \to 0$, so the loss is unbounded. For unclamped entropy maximization, $\nabla_{\boldsymbol{\theta}}(-H_t) \neq \mathbf{0}$ whenever the distribution is non-uniform, so the objective exerts gradient force even when H_t is close to $H_{\max}$ and forgetting is essentially complete. By contrast, clamped entropy produces *exactly zero* gradient once $H_t \geq \tau \log V$, eliminating wasted optimization pressure. □

Remark. Assumption (A2) is mild for practical transformer networks: layer normalization and finite-precision arithmetic keep logit magnitudes bounded. For Llama-2-7b with $V = 32000$ and typical logit ranges $B \approx 15$–30, the bound in (8) evaluates to approximately $(2 \cdot 30 + 2 \cdot 10.4) \cdot G \approx 81\,G$, a moderate constant. In contrast, the GA gradient $\|\nabla_{\boldsymbol{\theta}}(-\log p_{y_t})\|$ has no such guarantee as training drives $p_{y_t} \to 0$ on forget tokens.

A.2 Dual Variable Ratchet Convergence

BLADE enforces the retain-quality constraint $\mathcal{L}_{\text{ret}}(\boldsymbol{\theta}) \leq \varepsilon$ via an augmented Lagrangian with asymmetric dual updates. Let $r_t = \mathcal{L}_{\text{ret}}(\boldsymbol{\theta}_t) - \varepsilon$ denote the constraint residual at iteration t, $\rho > 0$ the penalty coefficient, $\alpha \in (0, 1)$ the asymmetry ratio (we use $\alpha = 0.1$), and $\lambda_{\min} > 0$ a dual variable floor. The dual variable is updated as:

$$\lambda_{t+1} = \max\Big(\lambda_{\min},\ \lambda_t + \begin{cases} \rho\, r_t & \text{if } r_t > 0, \\ \alpha\,\rho\, r_t & \text{if } r_t \leq 0. \end{cases}\Big) \tag{9}$$

We model training as proceeding through *episodes*, a simplifying abstraction motivated by the empirical observation that constraint violations cluster around epoch boundaries (see the remark below). Each episode k consists of a violation phase ($r_t > 0$ for n_k^+ consecutive steps with mean violation $\bar{r}_k^+ > 0$) followed by a recovery phase ($r_t \leq 0$ for n_k^- steps with mean slack $\bar{r}_k^- < 0$).

Proposition 2 (Dual Variable Ratchet). *Assume: (B1) the dual floor satisfies $\lambda_{\min} > 0$ (so $\lambda_t \geq \lambda_{\min}$ at all times); (B2) the asymmetry condition $n_k^+ \bar{r}_k^+ > \alpha\, n_k^- |\bar{r}_k^-|$ holds for each episode k; and (B3) there exists a dual threshold λ^* such that the ALM objective $\mathcal{L}_{\text{fgt}} + \lambda(\mathcal{L}_{\text{ret}} - \varepsilon) + \frac{\rho}{2}\max(0, r)^2$ admits a feasible stationary point with $r \leq 0$ whenever $\lambda \geq \lambda^*$. Then:*

(a) *The net dual change per episode satisfies $\Delta\lambda_k \geq \rho(n_k^+ \bar{r}_k^+ - \alpha\, n_k^- |\bar{r}_k^-|) > 0$.*

(b) *After at most $K^* = \lceil (\lambda^* - \lambda_0)/(\rho\,\underline{\Delta}) \rceil$ episodes, where $\underline{\Delta} = \min_k(n_k^+ \bar{r}_k^+ - \alpha\, n_k^- |\bar{r}_k^-|)$, the dual variable satisfies $\lambda_t \geq \lambda^*$ and no further violation episodes occur.*

Proof. (a) Since $\lambda_{\min} > 0$ and the violation phase increments λ by $\rho r_t > 0$ at each step, λ_t remains strictly above $\lambda_{\min}$ throughout the violation phase. During recovery, each decrement is $\alpha\rho|r_t|$, which is α times as large as the corresponding increment would be. The $\max(\lambda_{\min}, \cdot)$ projection can only *increase* λ_t relative to the unprojected update (it clips negative overshoots), so the cumulative change satisfies:

$$\begin{aligned} \Delta\lambda_k &= \lambda_{t_{k+1}} - \lambda_{t_k} \\ &\geq \rho \sum_{t \in \text{viol}} r_t + \alpha\rho \sum_{t \in \text{recov}} r_t \\ &= \rho\big(n_k^+ \bar{r}_k^+ - \alpha\, n_k^- |\bar{r}_k^-|\big) > 0, \end{aligned}$$

where the inequality uses the fact that $\max(\lambda_{\min}, x) \geq x$ for all x, so projecting can only reduce the total decrease. The final inequality follows from assumption (B2).

The asymmetry ratio $\alpha = 0.1$ makes (B2) easy to satisfy: even if the recovery phase is $10\times$ longer than the violation phase with equal magnitude, the condition holds with equality. In practice, violation spikes are sharp (large $\bar{r}_k^+$, small n_k^+) while recovery is gradual, so the condition is satisfied with margin.

(b) Since $\Delta\lambda_k \geq \rho\underline{\Delta} > 0$ for each episode, after K episodes:

$$\lambda_{t_K} \;\geq\; \lambda_0 + K\rho\underline{\Delta}.$$

Setting $K = K^*$ gives $\lambda_{t_{K^*}} \geq \lambda^*$. By assumption (B3), for $\lambda_t \geq \lambda^*$ the outer-loop optimizer can reach a point with $r_t \leq 0$, so no further violation phase begins. In the absence of violations, λ_t decreases slowly at rate $\alpha\rho|r_t|$ per step toward λ^* from above. $\square$

Remark on assumptions. Assumption (B1) corresponds to the hyperparameter $\lambda_{\min} = 0.1$ in our implementation, which prevents the dual variable from vanishing. Assumption (B3) is standard in ALM convergence theory (Bertsekas, 1999): it requires that the constrained problem is feasible for the given parameterization — i.e., there exist LoRA parameters achieving $\mathcal{L}_{\text{ret}} \leq \varepsilon$. The episode structure in (B2) is a modeling simplification; in practice, violations may be interleaved with recoveries, but the key property — that asymmetric updates cause net λ increase whenever total violation exceeds α times total slack — holds regardless of ordering.

Empirical validation. In our MUSE Books experiments, λ is initialized at 1.0 and spikes to ≈ 2.27 after a single violation episode at the epoch 1–2 boundary, then stabilizes. One episode with $\Delta\lambda_1 \approx 1.27$ suffices to cross λ^*, after which no further violations are observed across three subsequent epochs. A symmetric update ($\alpha = 1$) would decrease λ equally fast during recovery, risking oscillatory violations.

A.3 Selective Uncertainty via Entropy Clamping

We formalize the intuition that sub-maximal entropy clamping ($\tau < 1$) enables the model to be uncertain on forget data while remaining confident on retain data.

Proposition 3 (Selective Uncertainty). *Let $H_{\max} = \log V$ denote the maximum per-token entropy. Consider a model $p_{\boldsymbol{\theta}}$ trained with the clamped entropy objective (7) with threshold $\tau \in (0, 1)$.*

(a) *(Zero gradient above threshold.) For every forget token t with $H_t(\boldsymbol{\theta}) \geq \tau H_{\max}$, the gradient contribution $\nabla_{\boldsymbol{\theta}}\ell_t = \mathbf{0}$.*

(b) *(Gradient force at logit level for unclamped entropy.) For unclamped entropy maximization, $\mathcal{L}_{\text{ent}} = -\frac{1}{T}\sum_t H_t$, the logit-level gradient satisfies $\nabla_{z_t} H_t = \mathbf{0}$ if and only if $p_{\boldsymbol{\theta}}(\cdot \mid x_{<t})$ is uniform over V. Hence $\mathcal{L}_{\text{ent}}$ exerts a nonzero logit-level gradient on every token whose output distribution is non-uniform.*

(c) *(Decoupling at convergence.) At a stationary point of the combined objective $\mathcal{L}_{\text{fgt}} + \lambda\mathcal{L}_{\text{ret}}$ where all forget tokens satisfy $H_t > \tau H_{\max}$ (strict inequality), the stationarity condition reduces to $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\text{ret}} = \mathbf{0}$. That is, the retain loss alone governs the parameters, free from any opposing forget gradient.*

Proof. (a) Immediate from the definition: when $H_t \geq \tau H_{\max}$, the ReLU in (7) outputs zero, so $\ell_t = 0$ is constant in a neighborhood of $\boldsymbol{\theta}$, and $\nabla_{\boldsymbol{\theta}}\ell_t = \mathbf{0}$.

(b) The logit-level gradient is $\nabla_{z_t} H_t = -(\text{diag}(p) - pp^\top)(\log p + \mathbf{1})$. The softmax Jacobian $J = \text{diag}(p) - pp^\top$ has rank $V - 1$ (its null space is $\text{span}(\mathbf{1})$). Now $\nabla_{z_t} H_t = \mathbf{0}$ iff $J(\log p + \mathbf{1}) = \mathbf{0}$ iff $\log p + \mathbf{1} \in \text{span}(\mathbf{1})$ iff $\log p_v$ is constant for all v iff p is uniform. Therefore $\nabla_{z_t}(-H_t) \neq \mathbf{0}$ for every non-uniform distribution, confirming that unclamped entropy maximization pushes logits toward the constant-logit configuration $z_{t,v} = c$ for all v — which produces the uniform distribution.

Note that multiple parameter vectors $\boldsymbol{\theta}$ may produce uniform logits; the characterization is in logit space, not parameter space.

(c) If all forget tokens satisfy $H_t > \tau H_{\max}$ (strict inequality), then by continuity of $H_t(\boldsymbol{\theta})$ the strict inequality holds in a neighborhood of $\boldsymbol{\theta}$, so $\mathcal{L}_{\text{fgt}}(\boldsymbol{\theta}') = 0$ for all $\boldsymbol{\theta}'$ in this neighborhood and $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\text{fgt}} = \mathbf{0}$. The stationarity condition $\nabla_{\boldsymbol{\theta}}(\mathcal{L}_{\text{fgt}} + \lambda\mathcal{L}_{\text{ret}}) = \mathbf{0}$ becomes $\lambda\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\text{ret}} = \mathbf{0}$, and since $\lambda > 0$, this simplifies to $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\text{ret}} = \mathbf{0}$. Thus the parameters settle at a local minimum of the retain loss, with no opposing force from the forget objective. □

Information-theoretic interpretation. The clamping threshold partitions per-token entropy into two regimes: $[0, \tau H_{\max})$ where unlearning is active, and $[\tau H_{\max}, H_{\max}]$ where the forget objective is silent. With $\tau = 0.7$, forget tokens need only reach 70% of maximum entropy — uncertain enough that specific memorized content is not recoverable, but far from the uniform distribution that would require flattening all logits. The remaining 30% of entropy range can be viewed informally as a "confidence budget": shared parameters can maintain low-entropy retain predictions without the forget objective pushing them toward uniformity. This design is well-matched to LLM unlearning, where forget and retain data share most token-level statistics (common words, syntax) and differ only on knowledge-bearing tokens.

B Experimental Setup

B.1 Model and Data

For MUSE, we use Llama-2-7b-hf (Touvron et al., 2023). The MUSE benchmark (Shi et al., 2025) provides two corpora: **News** (BBC news articles) and **Books** (Harry Potter). For each corpus, MUSE supplies a *target model* (fine-tuned on the full corpus) and a *gold retrained model* (fine-tuned only on the retain set).

For TOFU (Maini et al., 2024), we use Llama-3.2-1B-Instruct and Llama-3.2-3B-Instruct. We evaluate on three forget splits: forget01 (1%, 20 QA pairs), forget05 (5%, 100 QA pairs), and forget10 (10%, 200 QA pairs). All TOFU results are averaged over 5 random seeds.

Table 14 summarizes the MUSE data splits. All sequences are chunked to a maximum length of 2048 tokens.

B.2 BLADE Hyperparameters

Table 15 lists the full hyperparameter configuration for BLADE on both MUSE corpora. The only differences between News and Books are the forget loss type (clamped entropy vs. logit margin), the ALM constraint threshold ε, and the number of training epochs.

	News	**Books**
Forget samples	889	554
Retain samples	1,777	1,108
Max sequence length	2,048	2,048
Steps/epoch (bs=2, accum=8)	~55	~34

Table 14: MUSE data statistics per corpus.

Hyperparameter	**News**	**Books**
LoRA		
Rank r	16	16
Alpha α	32	32
Dropout	0.0	0.0
Target modules	q, k, v, o, gate, up, down	
Bilevel optimization		
Inner steps K	3	3
Inner LR η_{in} (SGD)	2×10^{-4}	2×10^{-4}
Outer LR η_{θ} (Adam)	3×10^{-5}	3×10^{-5}
Gradient accumulation	8	8
Per-device batch size	2	2
Effective batch size	16	16
Max gradient norm	1.0	1.0
ALM constraint		
Retain threshold ε	0.70	0.70
Penalty ρ	0.1	0.1
Dual init λ_0	1.0	1.0
Dual floor $\lambda_{\min}$	0.1	0.1
Asymmetry ratio α	0.1	0.1
Forget loss		
Loss type	clamped_entropy	clamped_entropy
Clamping ratio τ	0.7	0.7
Training		
Epochs	4	4
Precision	bf16	bf16
Gradient checkpointing	✓	✓

Table 15: BLADE hyperparameters for MUSE News and Books.

B.3 LoRA Architecture

We apply LoRA adapters to all seven linear projections in each transformer layer: the four attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`) and the three feed-forward projections (`gate_proj`, `up_proj`, `down_proj`). With rank $r = 16$ and scaling $\alpha = 32$ (effective scaling factor $\alpha/r = 2.0$), this yields approximately 160M trainable parameters across 32 layers—roughly 2.3% of the full 6.7B model. The base model weights remain frozen; the reference model used for constraint evaluation is simply the base model with LoRA adapters disabled, incurring zero additional memory.

B.4 Baseline Configurations

All baselines use their default hyperparameters from the OpenUnlearning framework (Shi et al., 2025). Table 16 summarizes the key settings. We do not tune baseline hyperparameters; we use the configurations provided by the original authors.

Method	LR	Eff. BS	Epochs	Key params
GradAscent	1×10^{-5}	32	10	—
GradDiff	1×10^{-5}	32	10	$\gamma = 1.0$
NPO	3×10^{-5}	32	10	$\beta = 0.1$
SimNPO	1×10^{-5}	32	10	$\beta = 0.7$
BLURNPO	2.5×10^{-5}	32	10	$\beta = 0.05$
RMU	5×10^{-5}	32	—	layer 7, $\gamma_s = 2$
PDU	1×10^{-5}	32	10	$\alpha = 50$, $\varepsilon = 1.5$

Table 16: Baseline hyperparameters. All use AdamW with constant LR schedule. RMU targets specific layers rather than training for fixed epochs.

B.5 Evaluation Metrics

We evaluate using the standard MUSE evaluation suite. The five metrics are:

- **forget_knowmem** (↓): ROUGE-L F1 on knowledge QA pairs from the forget set (max 32 new tokens). Measures residual factual knowledge.
- **verbmem** (↓): ROUGE-L F1 on verbatim completion of forget-set passages (max 128 new tokens). Measures residual memorization.
- **retain** (↑): ROUGE-L F1 on knowledge QA pairs from the retain set. Measures preservation of non-targeted knowledge.
- **extract** (↓): Extraction strength—for each forget passage, the fraction of tokens the model can reproduce given a prefix. Lower means less recoverable memorization.
- **HM** (↑): Harmonic mean of (1 − forget_knowmem), (1 − verbmem), and retain. Penalizes imbalance across axes; any weak dimension tanks the score.

B.6 Compute

All experiments are run on a single NVIDIA H100 GPU (80GB). Training times are wall-clock (excluding evaluation). BLADE with clamped entropy on MUSE Books (4 epochs, 135 outer steps) takes approximately 104 minutes. Gradient checkpointing is enabled for all methods to fit within GPU memory.

C Extended Results

C.1 Full TOFU Results

Tables 17–18 report full TOFU results (mean±std over 5 seeds) for all methods and splits.

C.2 LLM Judge Results

Tables 19–20 report LLM judge scores (Claude Opus 4.7) on all TOFU splits and both model scales. All scores on 0–2 scale. HM_J = hmean(1−FL/2, RA/2, rRQ/2).

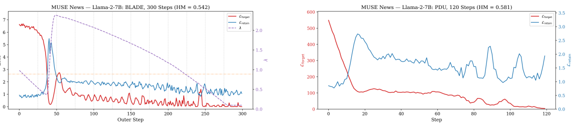
C.3 MUSE Books (Extended)

Table 21 reports MUSE Books results (5-fold for baselines where available; BLADE and BLURNPO seed=42 only).

LLM Judge. Table 22 reports LLM judge scores (5-fold for baselines; seed=42 for BLADE and BLURNPO).

C.4 MUSE News Dynamics

Figure 3 shows that BLADE's three-phase pattern holds on MUSE News, though the retain loss exhibits higher variance due to the heterogeneity of the 889-article corpus. Both BLADE and PDU show larger retain oscillations than on Books, but PDU's are uncontrolled while BLADE's are bounded by the ratcheted λ.



(a) BLADE (HM = 0.542) (b) PDU (HM = 0.581)

Figure 3: MUSE News dynamics. The distributed forget set produces higher retain variance for both methods. PDU achieves higher HM on this setting.

C.5 Training Dynamics

Table 23 shows the loss trajectory of BLADE on MUSE Books. The key pattern is a single retain spike at the epoch 1–2 boundary that ratchets λ from 1.0 to 2.27, after which all subsequent epochs run smoothly with no further violations.

Extended dynamics comparison across splits. Figure 4 shows BLADE dynamics on TOFU forget05 and forget10. As the forget set size increases, we observe: (i) the forget loss takes longer to converge (193 steps for forget05, 249 for forget10 vs. 110 for forget01), (ii) λ rises higher (∼3.5 for forget05, ∼4.5 for forget10) because more retain protection is

Table 17: TOFU (Llama-3.2-1B), 5 seeds. HM = hmean(MU, 1−Prob, 1−ROUGE). Best HM in **bold** (excl. Gold).

Split	Method	MU↑	Prob↓	ROUGE↓	HM↑
fgt01	Gold (retrain)	.597	.166	.414	.655
	GradAscent	.596±.001	.477±.010	.456±.009	.553±.006
	GradDiff	.581±.008	.421±.037	.464±.019	.564±.013
	NPO	.596±.001	.474±.010	.438±.015	.560±.008
	SimNPO	.594±.003	.858±.006	.734±.019	.240±.007
	RMU	.557±.002	.414±.008	.415±.004	.576±.002
	BLURNPO	.598±.003	.676±.013	.588±.018	.417±.010
	PDU	.602±.000	.186±.011	.313±.008	.690±.005
	BLADE	.599±.003	**.002**±.000	**.038**±.003	**.808**±.002
fgt05	Gold (retrain)	.599	.127	.383	.676
	GradAscent	.008±.013	.003±.005	.112±.083	.022±.037
	GradDiff	.453±.004	.073±.003	.375±.010	.614±.005
	NPO	.454±.013	.245±.006	.308±.006	.603±.007
	SimNPO	.596±.001	.848±.001	.741±.005	.248±.002
	RMU	.546±.001	.369±.006	.423±.007	.583±.003
	BLURNPO	.518±.019	.475±.074	.407±.052	.541±.037
	PDU	.588±.002	.073±.016	.214±.027	.740±.011
	BLADE	**.597**±.003	**.015**±.001	**.054**±.009	**.800**±.003
fgt10	Gold (retrain)	.591	.116	.379	.677
	GradAscent	.000±.000	.000±.000	.001±.001	.000±.000
	GradDiff	.435±.003	.047±.002	.339±.005	.617±.003
	NPO	.393±.025	.213±.005	.210±.010	.590±.020
	SimNPO	.597±.001	.842±.001	.734±.005	.255±.002
	RMU	.572±.002	.106±.015	.322±.012	.691±.008
	BLURNPO	.089±.138	.087±.040	.253±.085	.154±.198
	PDU	.592±.002	**.004**±.001	.065±.005	.797±.003
	BLADE	.593±.005	.009±.003	**.039**±.007	**.803**±.004

needed, and (iii) the retain loss exhibits mild oscillations on larger splits but stays bounded near ε. This adaptive scaling of λ with task difficulty requires no hyperparameter change.

PDU retain instability. Across all settings, PDU's retain loss exhibits a characteristic pattern: an initial spike followed by persistent oscillations that never fully resolve (Figure 4b,d). On forget05, PDU's retain loss fluctuates between 0.4 and 1.3 throughout training. On forget10, the oscillations are even larger (0.0–1.6). This instability arises because PDU's primal-dual updates are symmetric—the dual variable drops as quickly as it rises, providing no lasting protection after a retain violation. In contrast, BLADE's asymmetric 10× slower decay permanently records the severity of past violations.

C.6 Ablation Studies

Forget loss comparison. Table 24 compares forget loss variants within the BLADE framework on MUSE Books, all using the same bilevel ALM setup ($K = 3$, $\varepsilon = 0.15$, $\lambda_0 = 1.0$, $\rho = 0.1$). Each loss produces a qualitatively distinct training trajectory:

- **Logit margin:** Strong forgetting but unbounded gradients cause structural retain spikes at every epoch boundary. The ALM cannot fully recover because each spike's magnitude exceeds what K=3 inner steps can repair. Best retain ≈ 0.41.
- **Repr. orthogonal:** Completely stable retain (no spikes ever trigger) but weak forgetting—orthogonalizing hidden states does not suppress token-level generation.
- **Clamped entropy:** One spike that permanently ratchets λ, then smooth convergence. The bounded nature of the loss means subsequent outer steps are predictably sized, allowing the inner loop to fully repair retain damage. Best on both axes.

Complete component ablation. Table 25 reports the full component ablation on MUSE Books (seed=42). Each row modifies exactly one element from the full method (A0). We also report LLM judge scores to verify that automated metrics align with generation quality.

Key findings from the ablation:

- **A16 (ALM off):** The model collapses at

Table 18: TOFU (Llama-3.2-3B), 5 seeds. HM = hmean(MU, 1−Prob, 1−ROUGE). Best HM in **bold** (excl. Gold).

Split	Method	MU↑	Prob↓	ROUGE↓	HM↑
fgt01	Gold (retrain)	.663	.179	.409	.656
	GradAscent	$.668_{\pm.001}$	$.564_{\pm.009}$	$.525_{\pm.014}$	$.509_{\pm.008}$
	GradDiff	$.659_{\pm.003}$	$.569_{\pm.019}$	$.578_{\pm.027}$	$.483_{\pm.020}$
	NPO	$.668_{\pm.000}$	$.553_{\pm.009}$	$.494_{\pm.008}$	$.525_{\pm.005}$
	SimNPO	$.656_{\pm.002}$	$.920_{\pm.007}$	$.830_{\pm.021}$	$.151_{\pm.012}$
	RMU	$.657_{\pm.001}$	$.861_{\pm.002}$	$.715_{\pm.006}$	$.246_{\pm.003}$
	BLURNPO	$.667_{\pm.003}$	$.822_{\pm.015}$	$.788_{\pm.024}$	$.253_{\pm.021}$
	PDU	$\mathbf{.692}_{\pm.001}$	$.168_{\pm.006}$	$.283_{\pm.007}$	$.742_{\pm.003}$
	BLADE	$.654_{\pm.003}$	$\mathbf{.000}_{\pm.000}$	$\mathbf{.015}_{\pm.011}$	$\mathbf{.846}_{\pm.004}$
fgt05	Gold (retrain)	.659	.130	.388	.698
	GradAscent	$.481_{\pm.015}$	$.140_{\pm.015}$	$.333_{\pm.013}$	$.632_{\pm.008}$
	GradDiff	$.566_{\pm.005}$	$.147_{\pm.008}$	$.395_{\pm.010}$	$.653_{\pm.003}$
	NPO	$.542_{\pm.005}$	$.219_{\pm.004}$	$.359_{\pm.012}$	$.640_{\pm.003}$
	SimNPO	$.657_{\pm.001}$	$.901_{\pm.002}$	$.818_{\pm.005}$	$.175_{\pm.004}$
	RMU	$.644_{\pm.001}$	$.602_{\pm.004}$	$.534_{\pm.003}$	$.483_{\pm.002}$
	BLURNPO	$.640_{\pm.022}$	$.683_{\pm.082}$	$.593_{\pm.091}$	$.412_{\pm.075}$
	PDU	$\mathbf{.687}_{\pm.001}$	$.009_{\pm.001}$	$.085_{\pm.009}$	$\mathbf{.843}_{\pm.002}$
	BLADE	$.655_{\pm.002}$	$\mathbf{.007}_{\pm.006}$	$\mathbf{.028}_{\pm.016}$	$.842_{\pm.005}$
fgt10	Gold (retrain)	.661	.115	.382	.698
	GradAscent	$.000_{\pm.000}$	$.000_{\pm.000}$	$.003_{\pm.004}$	$.000_{\pm.000}$
	GradDiff	$.532_{\pm.015}$	$.079_{\pm.008}$	$.361_{\pm.024}$	$.662_{\pm.005}$
	NPO	$.549_{\pm.013}$	$.237_{\pm.005}$	$.353_{\pm.059}$	$.640_{\pm.015}$
	SimNPO	$.652_{\pm.001}$	$.889_{\pm.002}$	$.808_{\pm.004}$	$.191_{\pm.003}$
	RMU	$.647_{\pm.001}$	$.390_{\pm.010}$	$.472_{\pm.002}$	$.591_{\pm.004}$
	BLURNPO	$.385_{\pm.167}$	$.245_{\pm.163}$	$.314_{\pm.094}$	$.502_{\pm.099}$
	PDU	$\mathbf{.680}_{\pm.009}$	$\mathbf{.000}_{\pm.000}$	$\mathbf{.029}_{\pm.008}$	$\mathbf{.857}_{\pm.003}$
	BLADE	$.647_{\pm.004}$	$.005_{\pm.003}$	$.038_{\pm.014}$	$.836_{\pm.006}$

step 31 (L_{ret}=10.56 triggered safety break). Pure clamped entropy without constraint enforcement destroys the model. The ALM is the core stabilizing mechanism.

- **A15 (No LoRA):** Full fine-tuning achieves aggressive forgetting (fk=0.002) but catastrophic retain collapse (ret=0.306, −0.352). The low-rank bottleneck provides essential implicit regularization that the bilevel structure alone cannot replace.
- **A9 (τ=1.0):** Without clamping, gradients continue pushing already-forgotten tokens, causing 8.2% retain degradation (0.658→0.604). The per-token stopping criterion is essential for protecting shared representations.
- **A1 (K=0):** Modest retain drop (−0.027). The ALM constraint alone provides strong protection, but the inner loop adds a safety margin by pre-repairing retain before each outer step. Training is 2.5× faster per step without the inner loop. Figure 5 visualizes the mechanism: during the phase-transition spike (steps 28–40), the inner loop fires 18 emergency SGD steps that pull L_{ret} down faster, resulting in 1.3 less cumulative constraint violation (17.5 vs. 18.8). This 7% violation reduction translates directly to the 0.027 retain gap at convergence.
- **A18 (Swapped):** Putting forgetting in the inner loop causes λ to explode to 367 because the constraint threshold ε_{fgt}=22.4 exceeds clamped entropy's theoretical maximum ($\tau \cdot H_{\max} \approx 8.3$)—physically impossible to satisfy.
- **A4/A17 (Unbounded losses):** GA produces L_{fgt}=−7.2 (unbounded negative) and collapses retain; logit margin destroys all representations even with full ALM protection. Both confirm that bounded forget losses are essential for bilevel stability.
- **A5 (NPO):** The NPO loss collapses to 0.0 by step 40 without actually unlearning—verbatim memorization remains at 0.997. The model satisfies NPO trivially because long-sequence probabilities under the reference model are already low, regardless of memorization.

Table 19: LLM Judge on TOFU 1B (5 seeds). Best $\mathrm{HM_J}$ in **bold**.

Split	Method	FL↓	RA↑	rRQ↑	$\mathrm{HM_J}$↑
fgt01	GradAscent	1.105$_{\pm.051}$	1.660$_{\pm.008}$	1.991$_{\pm.001}$	.674$_{\pm.019}$
	GradDiff	1.205$_{\pm.068}$	1.541$_{\pm.035}$	1.972$_{\pm.006}$	.620$_{\pm.024}$
	NPO	1.050$_{\pm.076}$	1.673$_{\pm.007}$	1.993$_{\pm.001}$	.696$_{\pm.028}$
	SimNPO	1.755$_{\pm.033}$	1.655$_{\pm.007}$	1.986$_{\pm.002}$	.288$_{\pm.030}$
	RMU	0.814$_{\pm.051}$	1.392$_{\pm.011}$	1.929$_{\pm.006}$	.721$_{\pm.013}$
	BLURNPO	1.400$_{\pm.047}$	1.667$_{\pm.009}$	1.992$_{\pm.002}$	.541$_{\pm.025}$
	PDU	0.855$_{\pm.060}$	1.597$_{\pm.011}$	1.966$_{\pm.003}$	.746$_{\pm.017}$
	BLADE	**0.015**$_{\pm.020}$	1.654$_{\pm.010}$	1.977$_{\pm.005}$	**.929**$_{\pm.005}$
fgt05	GradAscent	0.074$_{\pm.065}$	0.101$_{\pm.088}$	0.510$_{\pm.425}$	.117$_{\pm.101}$
	GradDiff	0.651$_{\pm.027}$	1.006$_{\pm.014}$	1.678$_{\pm.011}$	.643$_{\pm.004}$
	NPO	0.440$_{\pm.018}$	1.139$_{\pm.046}$	1.885$_{\pm.020}$	.731$_{\pm.014}$
	SimNPO	1.575$_{\pm.010}$	1.693$_{\pm.014}$	1.987$_{\pm.003}$	.435$_{\pm.007}$
	RMU	0.719$_{\pm.042}$	1.363$_{\pm.018}$	1.910$_{\pm.011}$	.736$_{\pm.008}$
	BLURNPO	0.858$_{\pm.075}$	1.370$_{\pm.063}$	1.982$_{\pm.014}$	.709$_{\pm.007}$
	PDU	0.368$_{\pm.075}$	1.559$_{\pm.015}$	1.959$_{\pm.006}$	.850$_{\pm.013}$
	BLADE	**0.067**$_{\pm.023}$	1.643$_{\pm.015}$	1.978$_{\pm.007}$	**.919**$_{\pm.002}$
fgt10	GradAscent	0.000$_{\pm.000}$	0.000$_{\pm.000}$	0.000$_{\pm.000}$	.000$_{\pm.000}$
	GradDiff	0.461$_{\pm.008}$	0.912$_{\pm.018}$	1.528$_{\pm.035}$	.625$_{\pm.007}$
	NPO	0.443$_{\pm.019}$	0.943$_{\pm.067}$	1.821$_{\pm.041}$	.665$_{\pm.025}$
	SimNPO	1.576$_{\pm.023}$	1.715$_{\pm.018}$	1.985$_{\pm.002}$	.435$_{\pm.017}$
	RMU	0.285$_{\pm.042}$	1.498$_{\pm.013}$	1.976$_{\pm.004}$	.854$_{\pm.010}$
	BLURNPO	0.262$_{\pm.244}$	0.357$_{\pm.378}$	0.727$_{\pm.636}$	.255$_{\pm.213}$
	PDU	0.042$_{\pm.009}$	1.567$_{\pm.010}$	1.976$_{\pm.005}$	.906$_{\pm.002}$
	BLADE	0.043$_{\pm.022}$	1.626$_{\pm.018}$	1.976$_{\pm.004}$	**.919**$_{\pm.005}$

C.7 KnowUndo Extended Results

Table 26 reports the full KnowUndo results with all metrics.

Analysis. KnowUndo evaluates a qualitatively different setting: the target model is LoRA-fine-tuned (not fully fine-tuned), so memorization is concentrated in the adapter weights. BLADE adapts naturally by initializing from the target LoRA checkpoint and unlearning on top. The privacy domain has 2.7× higher initial memorization than copyright (forget_ROUGE 0.665 vs. 0.244), making it a harder target. BLADE's advantage is most pronounced on privacy (+0.049 HM over PDU), suggesting that the bilevel structure handles high-memorization settings particularly well—the stronger the memorization, the more benefit from bounded forgetting and adaptive constraint enforcement.

C.8 MUSE News Extended Results

MUSE News targets removal of 889 BBC news articles from Llama-2-7B. Unlike MUSE Books (a single cohesive narrative), News consists of short factual articles—a distributed forget set where knowledge is spread across many independent documents. Table 27 reports 5-fold results (seed=42 for BLADE and BLURNPO due to OOM constraints).

LLM Judge. Table 28 reports LLM judge scores (5-fold for baselines; seed=42 for BLADE and BLURNPO).

Why PDU wins on News. The distributed forget set creates heterogeneous memorization levels across articles. PDU's logit-margin loss operates on individual token predictions and naturally adapts to per-sample difficulty. BLADE's clamped entropy, while bounded and self-stabilizing, uses a single global λ to balance 889 articles with varying memorization—articles that forget quickly contribute zero gradient (good), but the remaining stubborn articles face a λ calibrated to the ensemble rather than their individual difficulty. This suggests that per-document or per-cluster constraint enforcement could improve BLADE on distributed settings.

C.9 Sustainability and Scalability

We evaluate PDU and BLADE on MUSE News for two additional MUSE properties: *scalability* (performance on progressively larger forget sets in a single run) and *sustainability* (performance under sequential unlearning requests).

PDU maintains reasonable performance at 1–3× forget scale (HM≈0.58) but collapses

Table 20: LLM Judge on TOFU 3B (5 seeds). Best HM_J in **bold**.

Split	Method	FL↓	RA↑	rRQ↑	HM_J↑
fgt01	GradAscent	$1.335_{\pm.086}$	$1.849_{\pm.003}$	$1.998_{\pm.000}$	$.587_{\pm.045}$
	GradDiff	$1.500_{\pm.066}$	$1.818_{\pm.017}$	$1.997_{\pm.001}$	$.490_{\pm.041}$
	NPO	$1.230_{\pm.041}$	$1.843_{\pm.004}$	$1.998_{\pm.001}$	$.640_{\pm.019}$
	SimNPO	$1.825_{\pm.053}$	$1.839_{\pm.008}$	$1.995_{\pm.001}$	$.220_{\pm.057}$
	RMU	$1.585_{\pm.038}$	$1.807_{\pm.002}$	$1.995_{\pm.001}$	$.432_{\pm.027}$
	BLURNPO	$1.730_{\pm.122}$	$1.853_{\pm.003}$	$1.998_{\pm.001}$	$.308_{\pm.109}$
	PDU	$0.615_{\pm.042}$	$1.722_{\pm.007}$	$1.968_{\pm.004}$	$.828_{\pm.011}$
	BLADE	$\mathbf{0.010}_{\pm.014}$	$1.845_{\pm.004}$	$1.995_{\pm.001}$	$\mathbf{.970}_{\pm.002}$
fgt05	GradAscent	$0.693_{\pm.042}$	$1.452_{\pm.021}$	$1.981_{\pm.012}$	$.766_{\pm.009}$
	GradDiff	$0.833_{\pm.007}$	$1.293_{\pm.022}$	$1.705_{\pm.062}$	$.677_{\pm.009}$
	NPO	$0.738_{\pm.037}$	$1.552_{\pm.009}$	$1.995_{\pm.001}$	$.774_{\pm.009}$
	SimNPO	$1.742_{\pm.013}$	$1.854_{\pm.007}$	$1.996_{\pm.001}$	$.305_{\pm.012}$
	RMU	$1.107_{\pm.033}$	$1.703_{\pm.008}$	$1.991_{\pm.003}$	$.679_{\pm.014}$
	BLURNPO	$1.397_{\pm.153}$	$1.712_{\pm.054}$	$1.997_{\pm.001}$	$.538_{\pm.078}$
	PDU	$0.067_{\pm.013}$	$1.749_{\pm.009}$	$1.978_{\pm.003}$	$.941_{\pm.002}$
	BLADE	$\mathbf{0.023}_{\pm.026}$	$1.817_{\pm.011}$	$1.990_{\pm.004}$	$\mathbf{.962}_{\pm.004}$
fgt10	GradAscent	$0.000_{\pm.000}$	$0.000_{\pm.000}$	$0.000_{\pm.000}$	$.000_{\pm.000}$
	GradDiff	$0.663_{\pm.029}$	$1.164_{\pm.058}$	$1.432_{\pm.199}$	$.648_{\pm.037}$
	NPO	$0.613_{\pm.078}$	$1.519_{\pm.042}$	$1.992_{\pm.006}$	$.796_{\pm.011}$
	SimNPO	$1.713_{\pm.006}$	$1.858_{\pm.005}$	$1.995_{\pm.002}$	$.331_{\pm.006}$
	RMU	$0.851_{\pm.017}$	$1.703_{\pm.006}$	$1.989_{\pm.002}$	$.765_{\pm.005}$
	BLURNPO	$0.726_{\pm.354}$	$1.046_{\pm.508}$	$1.508_{\pm.601}$	$.550_{\pm.161}$
	PDU	$0.014_{\pm.005}$	$1.726_{\pm.029}$	$1.952_{\pm.038}$	$.940_{\pm.011}$
	BLADE	$0.034_{\pm.028}$	$1.837_{\pm.011}$	$1.992_{\pm.003}$	$\mathbf{.965}_{\pm.004}$

Method	**fk↓**	**vm↓**	**ret↑**	**HM↑**
Gold (retrain)	0.303	0.145	0.687	0.739
Target (pre-unlearn)	0.471	0.997	0.691	0.009
GradAscent	$0.000_{\pm.00}$	$0.000_{\pm.00}$	$0.000_{\pm.00}$	$0.000_{\pm.00}$
GradDiff	$0.000_{\pm.00}$	$0.000_{\pm.00}$	$0.004_{\pm.00}$	$0.011_{\pm.01}$
NPO	$0.303_{\pm.02}$	$0.342_{\pm.03}$	$0.574_{\pm.02}$	$0.638_{\pm.01}$
SimNPO	$0.238_{\pm.02}$	$0.002_{\pm.00}$	$0.600_{\pm.01}$	$0.753_{\pm.01}$
RMU	$0.210_{\pm.00}$	$0.113_{\pm.01}$	$0.598_{\pm.01}$	$0.738_{\pm.00}$
PDU	$0.139_{\pm.01}$	$0.129_{\pm.01}$	$0.372_{\pm.01}$	$0.600_{\pm.01}$
BLURNPO (seed=42)	0.175	0.000	0.527	0.730
BLADE (seed=42)	**0.110**	**0.000**	**0.658**	**0.823**

Table 21: MUSE Books results (Llama-2-7b, 5-fold). **Bold**: best among unlearning methods.

Method	**FL↓**	**RA↑**	**rRQ↑**	$\mathbf{HM_J}$**↑**
GradAscent	$0.00_{\pm.00}$	$0.00_{\pm.00}$	$0.00_{\pm.00}$	$.000_{\pm.000}$
GradDiff	$0.00_{\pm.00}$	$0.04_{\pm.02}$	$0.01_{\pm.01}$	$.010_{\pm.009}$
NPO	$0.91_{\pm.03}$	$1.28_{\pm.03}$	$1.61_{\pm.01}$	$.647_{\pm.008}$
SimNPO	$0.31_{\pm.12}$	$1.28_{\pm.02}$	$1.62_{\pm.01}$	$.753_{\pm.020}$
RMU	$0.32_{\pm.02}$	$1.41_{\pm.02}$	$1.68_{\pm.01}$	$.791_{\pm.004}$
PDU	$0.53_{\pm.02}$	$0.99_{\pm.02}$	$1.15_{\pm.02}$	$.586_{\pm.007}$
BLURNPO (seed=42)	0.21	1.09	1.47	.696
BLADE (seed=42)	**0.14**	**1.40**	**1.69**	**.814**

Table 22: MUSE Books LLM judge (5-fold). HM_J = hmean(1−FL/2, RA/2, rRQ/2).

at 4× (HM=0.005). For sustainability, PDU survives 2 sequential unlearning steps (HM 0.580→0.558) but is destroyed by step 3 (HM=0.016). The LLM judge confirms: step 2 produces coherent outputs (HM_J=0.680) but steps 3–4 generate incoherent text (RA=0, RQ=0).

In contrast, BLADE shows stable scalability (HM 0.542→0.547→0.552 at 1–3× scale) and sustainability (HM 0.542→0.545→0.523 over 3 sequential steps). While PDU achieves higher HM on a single standard-size forget set, BLADE's bounded loss and LoRA parameterization prevent the catastrophic retain collapse that PDU suffers under repeated or scaled application. This suggests that BLADE's architecture is better suited for practical deployment where multiple unlearning requests arrive over time.

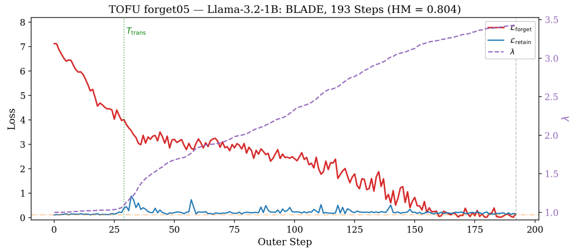
D Training Dynamics Analysis

We analyze the loss curve behavior of BLADE and PDU across benchmarks and identify structural patterns that explain the performance differences observed in Tables 17–29.

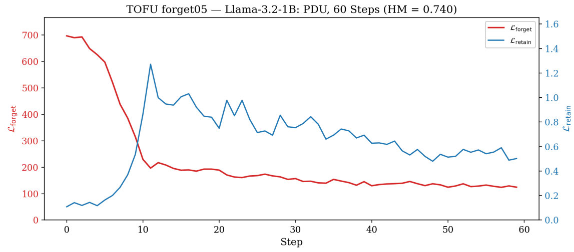
D.1 Three-Phase Dynamics of BLADE

Across all benchmarks and forget set sizes, BLADE exhibits a universal three-phase training pattern:

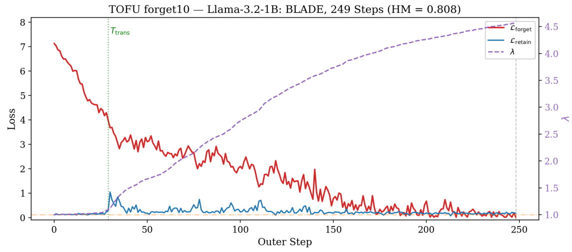
Phase 1: Warm-up (steps 1–T_{trans}). The forget loss decreases steadily from its initial value (∼7.0 for MUSE Books) while retain loss stays near ε. The dual variable λ remains approximately constant at its initial value ($\lambda_0 =$



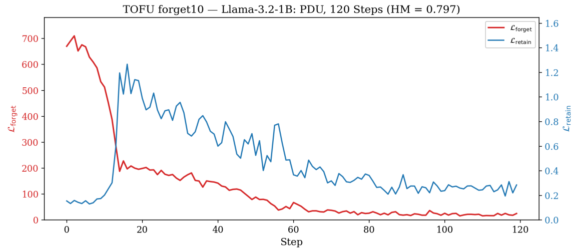
(a) TOFU 1B forget05: BLADE, 193 steps (HM = 0.804)



(b) TOFU 1B forget05: PDU, 60 steps (HM = 0.740)



(c) TOFU 1B forget10: BLADE, 249 steps (HM = 0.808)



(d) TOFU 1B forget10: PDU, 120 steps (HM = 0.797)

Figure 4: Training dynamics on larger forget splits. BLADE's λ adapts to task difficulty (higher for larger forget sets) while PDU shows persistent retain oscillations on all settings.

Step	$\mathcal{L}_{\text{fgt}}$	$\mathcal{L}_{\text{ret}}$	λ	r	Notes
0	6.995	0.052	1.00	−0.10	Init
30	5.861	0.153	0.97	+0.00	First violation
33	4.275	1.167	1.05	+1.02	Epoch 1→2 spike
36	2.186	1.649	1.41	+1.50	Spike peak
48	0.910	0.276	2.24	+0.13	ALM recovering
66	0.227	0.054	2.26	−0.10	Epoch 2 end
72	0.073	0.051	2.26	−0.10	Epoch 3 start, no spike
102	0.015	0.047	2.22	−0.10	Epoch 4 start, no spike
135	0.008	0.037	2.19	−0.11	Converged

Table 23: Training dynamics of BLADE on MUSE Books (4 epochs, 135 steps). $r = \mathcal{L}_{\text{ret}} - \varepsilon$ with $\varepsilon = 0.15$.

ID	Forget loss	Ep.	fk↓	vm↓	ret↑	HM↑
13	logit_margin	2	0.093	0.001	0.389	0.564
14	focal_logit_margin	2	0.112	0.001	0.391	0.567
16	repr_orthogonal	2	0.357	0.313	0.618	0.648
17	clamped_entropy	2	0.096	0.001	0.443	0.687
18	**clamped_entropy**	**4**	**0.080**	**0.000**	**0.598**	**0.798**

Table 24: Forget loss ablation within BLADE on MUSE Books. All use the same bilevel ALM configuration. Clamped entropy with 4 epochs achieves the best trade-off.

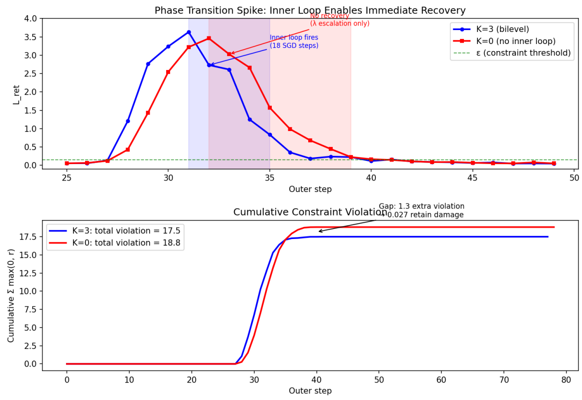


Figure 5: Phase transition spike: inner loop (K=3, blue) vs. no inner loop (K=0, red). Top: $\mathcal{L}_{\text{ret}}$ during the spike episode. The inner loop fires emergency SGD steps (blue shading) that pull retain down faster. Bottom: cumulative constraint violation. K=3 accumulates 7% less total violation (17.5 vs. 18.8), explaining the 0.027 retain gap at convergence.

1.0). During this phase, the clamped entropy objective is "easy"—most tokens have low entropy and contribute gradient—so the outer loop makes rapid progress without disturbing retain.

Phase 2: Spike-and-ratchet (T_{trans} to $T_{\text{trans}} + {\sim}15$ steps). At a critical transition point T_{trans}, the model encounters a phase transition where the forget loss drops sharply but temporarily destabilizes retain. The retain loss spikes (e.g., from 0.15 to 1.65 on MUSE Books), triggering the asymmetric dual update which rapidly ratchets λ upward ($1.0 \rightarrow 2.27$ on Books). The inner loop fires K emergency SGD steps per outer step to repair retain dam-

Table 25: Complete ablation on MUSE Books (seed=42). Each row changes one component from the full BLADE (A0). LLM judge: FL = forget leakage, RA = retain accuracy, ret_RQ = retain response quality (all 0–2 scale).

ID	Configuration	fk↓	vm↓	ret↑	HM↑	Δret	FL_J↓	RA_J↑	rRQ_J↑	HM_J↑
A0	Full BLADE	.110	.000	.658	**.823**	—	0.14	1.40	1.69	.814
A1	$K=0$ (no inner loop)	.072	.000	.631	.819	−.027	0.10	1.37	1.68	.811
A9	$\tau=1.0$ (unclamped)	.161	.003	.604	.780	−.054	0.20	1.34	1.64	.785
A15	No LoRA (full FT)	.002	.000	.306	.459	−.352	0.00	0.70	1.25	.550
A16	ALM off ($\lambda=0$, $\rho=0$)	.000	.002	.092	.233	−.566	0.01	0.24	0.28	.117
A18	Swapped bilevel	.372	.643	.651	.506	−.007	—	—	—	—
A4	GA forget loss	.006	.003	.060	.160	−.598	0.01	0.10	0.18	.093
A5	NPO forget loss	.457	.997	.662	.009	+.004	1.44	1.47	1.72	.495
A17	Logit margin loss	.000	.000	.000	.000	−.658	0.00	0.00	0.00	.000

Table 26: KnowUndo full results (Llama-2-7B-chat, seed=42). HM = hmean(1−fgt_ROUGE, ret_ROUGE, MMLU). LLM judge scores on 0–1 scale (Claude Sonnet 4.6).

	Copyright						LLM Judge		
Method	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	RQ↑
FT (target)	.854	.244	.894	.336	.439	.456	.602	.830	.795
GradDiff	.000	.000	.623	.200	.387	.349	.000	.344	.239
NPO	.670	.193	.757	.300	.447	.441	.243	.783	.740
SimNPO	.245	.012	.786	.318	.435	.464	.000	.755	.556
RMU	.444	.081	.768	.207	.390	.353	.000	.453	.479
PDU	.051	.007	.846	.302	.441	.456	.007	.748	.500
BLADE	**.133**	**.009**	**.848**	**.334**	.442	**.479**	.021	**.875**	**.612**

	Privacy						LLM Judge		
Method	fgt_Acc↓	fgt_R↓	ret_Acc↑	ret_R↑	MMLU↑	HM↑	FL↓	RA↑	RQ↑
FT (target)	.940	.665	.949	.698	.445	.450	.882	.847	.975
GradDiff	.070	.056	.753	.427	.434	.526	.028	.495	.379
NPO	.634	.313	.786	.431	.449	.500	.282	.555	.897
SimNPO	.486	.289	.889	.540	.441	.543	.177	.657	.548
RMU	.578	.072	.669	.084	.361	.190	.009	.014	.081
PDU	.059	.032	.809	.468	.441	.552	.018	.574	.379
BLADE	**.209**	**.075**	**.920**	**.621**	.435	**.601**	.068	**.732**	**.477**

Method	fk↓	vm↓	rk↑	HM↑
Gold	.324	.204	.552	.660
GradDiff	$.329_{\pm.03}$	$.043_{\pm.02}$	$.269_{\pm.02}$	$.479_{\pm.02}$
NPO	$.517_{\pm.02}$	$.274_{\pm.02}$	$.435_{\pm.01}$	$.522_{\pm.01}$
SimNPO	$.628_{\pm.01}$	$.542_{\pm.01}$	$.513_{\pm.01}$	$.440_{\pm.00}$
RMU	$.495_{\pm.01}$	$.267_{\pm.01}$	$.434_{\pm.01}$	$.531_{\pm.00}$
PDU	$.525_{\pm.02}$	$.092_{\pm.07}$	$.508_{\pm.04}$	$\mathbf{.577}_{\pm.01}$
BLURNPO (seed=42)	.317	.181	.304	.448
BLADE (seed=42)	.550	.173	.475	.542

Table 27: MUSE News (5-fold, seeds=42/123/456/789/1024). BLURNPO OOM on 4×40GB.

Method	**FL↓**	**RA↑**	**rRQ↑**	**HM_J↑**
GradDiff	$0.58_{\pm.02}$	$0.79_{\pm.04}$	$1.16_{\pm.14}$	$.529_{\pm.030}$
NPO	$1.20_{\pm.03}$	$1.05_{\pm.02}$	$1.58_{\pm.01}$	$.531_{\pm.011}$
SimNPO	$1.49_{\pm.01}$	$1.24_{\pm.01}$	$1.72_{\pm.01}$	$.447_{\pm.007}$
RMU	$1.13_{\pm.01}$	$1.15_{\pm.02}$	$1.60_{\pm.01}$	$.565_{\pm.004}$
PDU	$0.87_{\pm.12}$	$1.18_{\pm.05}$	$1.64_{\pm.05}$	$\mathbf{.638}_{\pm.019}$
BLURNPO (seed=42)	0.47	0.51	1.47	.455
BLADE (seed=42)	0.85	0.88	1.72	.579

Table 28: MUSE News LLM judge (5-fold). HM_J = hmean(1−FL/2, RA/2, rRQ/2).

age. This phase is brief but critical: it permanently calibrates λ to the task difficulty.

Phase 3: Smooth convergence ($T_{\mathbf{trans}}+15$ to end). After the ratchet, the elevated λ provides sufficient retain protection. The forget loss continues to decrease but with a shrinking effective gradient (fewer tokens remain "active" above the clamp threshold). Retain loss stays stably below ε. The λ value slowly decays by the asymmetric factor (0.1ρ per step of satisfaction) but never returns to pre-ratchet levels.

Stability of $T_{\mathbf{trans}}$. The transition point $T_{\text{trans}} \approx 28$ steps is remarkably stable across benchmarks: 28 on MUSE Books, 30 on TOFU

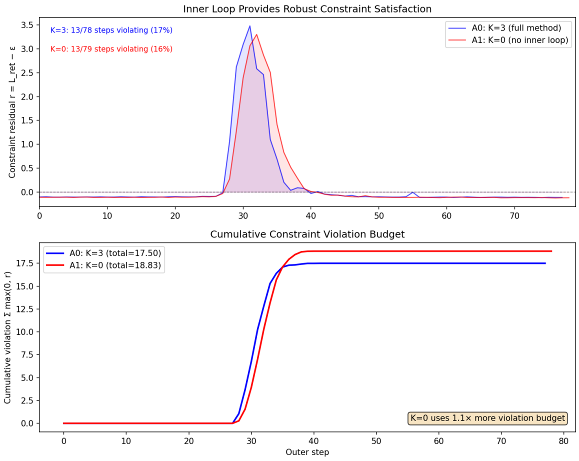


Figure 6: Constraint residual $r = L_{\text{ret}} - \varepsilon$ over training. Both K=3 and K=0 have the same number of violating steps (13), but K=0's violations extend later in training (wider red shading), producing slightly more cumulative damage. After step 50, both methods achieve permanent constraint satisfaction.

1B fgt01, 28 on TOFU 1B fgt05. This stability arises because T_{trans} is determined by when the LoRA adapter has accumulated enough low-rank perturbation to qualitatively change the output distribution on forget tokens—a property of the LoRA parameterization rather than the dataset.

D.2 Adaptive λ as Difficulty Diagnostic

The final λ value after the ratchet phase scales monotonically with task difficulty:

- TOFU 1B fgt01 (20 QA pairs): $\lambda_{\text{final}} \approx 2.4$
- TOFU 1B fgt05 (100 QA pairs): $\lambda_{\text{final}} \approx 3.6$
- TOFU 1B fgt10 (200 QA pairs): $\lambda_{\text{final}} \approx 4.5$
- MUSE Books (554 documents): $\lambda_{\text{final}} \approx 2.3$

Larger forget sets require more aggressive forgetting, which causes larger retain spikes, which in turn ratchet λ higher. This automatic scaling requires no hyperparameter adjustment between settings—the same initial $\lambda_0 = 1.0$ and $\rho = 0.1$ produce appropriate constraint enforcement across a 10× range of forget set sizes.

λ plateau as convergence diagnostic. Once λ reaches its plateau (visible as a flat segment in the dynamics plots), the system has found a stable balance between forgetting and retention. This provides a *built-in convergence diagnostic* without requiring held-out validation: if λ is still rising, the model has not yet found an equilibrium; if λ is flat, training can be safely stopped. In practice, λ plateaus 10–15 steps after the spike, coinciding with $\mathcal{L}_{\text{fgt}}$ dropping below 1.0.

Scalability (single-shot, increasing forget size)					
Scale	Method	fk↓	vm↓	rk↑	HM↑
1× (889)	PDU	.542	.065	.522	.581
	BLADE	.550	.173	.475	.542
2× (1778)	PDU	.495	.050	.467	.579
	BLADE	.504	.321	.500	.547
3× (2667)	PDU	.538	.024	.488	.573
	BLADE	.466	.366	.505	.552
4× (3554)	PDU	.321	.021	.002	.005
	BLADE		(running)		

Sustainability (sequential unlearning steps)					
Step	Method	fk↓	vm↓	rk↑	HM↑
1	PDU	.542	.065	.522	.580
	BLADE	.550	.173	.475	.542
2	PDU	.554	.007	.470	.558
	BLADE	.539	.212	.485	.545
3	PDU	.550	.001	.005	.016
	BLADE	.556	.230	.459	.523
4	PDU	.591	.003	.051	.130
	BLADE		(running)		

Table 29: Scalability and sustainability on MUSE News. PDU collapses at 4× scale and after step 2; BLADE maintains stable HM through step 3.

D.3 PDU Instability: Oscillatory Limit Cycles

PDU (Entesari et al., 2025) uses an unbounded logit-margin loss with symmetric primal-dual updates. Our dynamics analysis reveals a structural instability:

Unbounded loss ⇒ divergent gradients. PDU's forget loss (logit margin) has no upper bound. As training progresses and the model becomes more confident on forget tokens, the margin loss grows, producing increasingly large outer perturbations. Unlike BLADE's clamped entropy (bounded by $\tau \cdot H_{\max}$), there is no mechanism to attenuate the forget gradient as forgetting succeeds.

Symmetric dual ⇒ no ratchet. PDU's dual variable uses symmetric updates: it rises during violation and falls equally fast during satisfaction. After a retain spike ratchets the dual up, subsequent satisfaction immediately erodes the protection. This creates a limit cycle: large perturbation → retain spike → λ rises → retain recovers → λ falls → large perturbation again.

Empirical signature. On MUSE News, PDU's retain loss oscillates between 0.8 and 3.0 throughout training, never achieving the stable sub-ε convergence that BLADE exhibits after its single ratchet event. On MUSE Books, PDU shows 5+ distinct retain spikes across 4 epochs, each partially repaired but never fully resolved. These oscillations directly cause the worse HM scores: each spike damages shared representations, and while the token-level metrics may recover, the generation quality (measured by LLM judge) does not fully bounce back.

Implication for scalability. The oscillatory instability compounds under repeated application (sustainability) and larger forget sets (scalability). Each sequential unlearning step starts from a model that has accumulated representational damage from prior oscillations. By step 3, PDU's retain loss diverges entirely (HM=0.016). BLADE's monotonic λ ratchet prevents this accumulation: each application starts from a model where λ has permanently encoded the protection level needed.

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